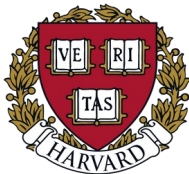




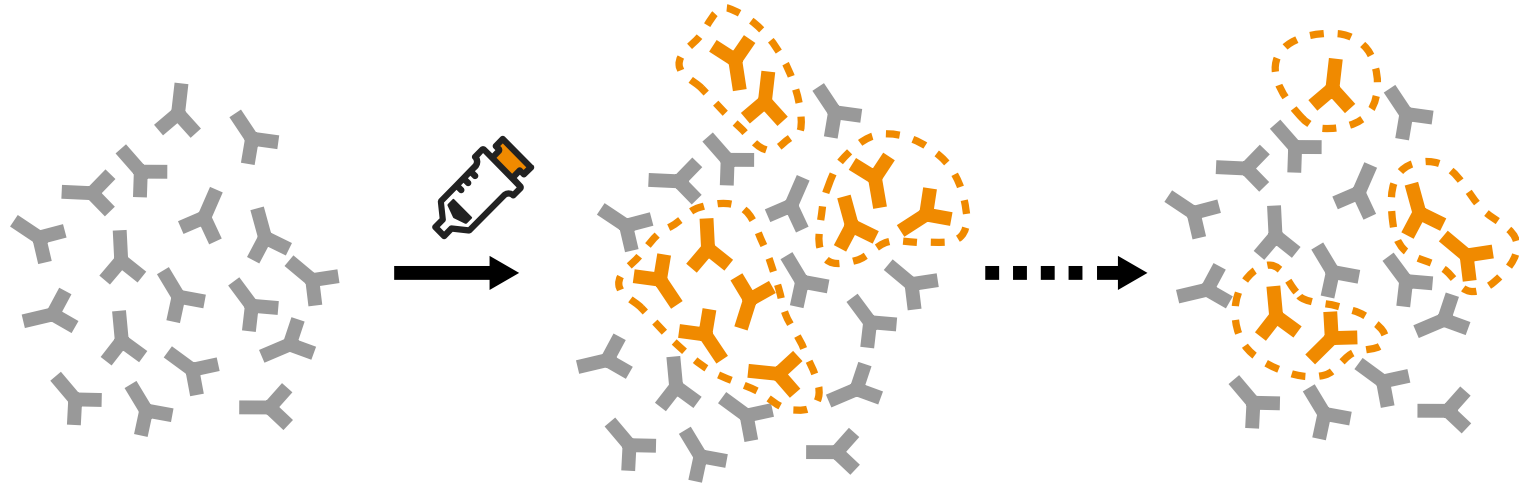
Naive Binding Affinity Shapes Clonal Selection and Evolution in the B-cell Response to SARS-CoV-2 vaccination

Thomas Duplic

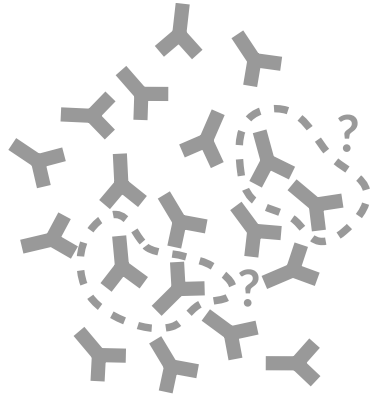
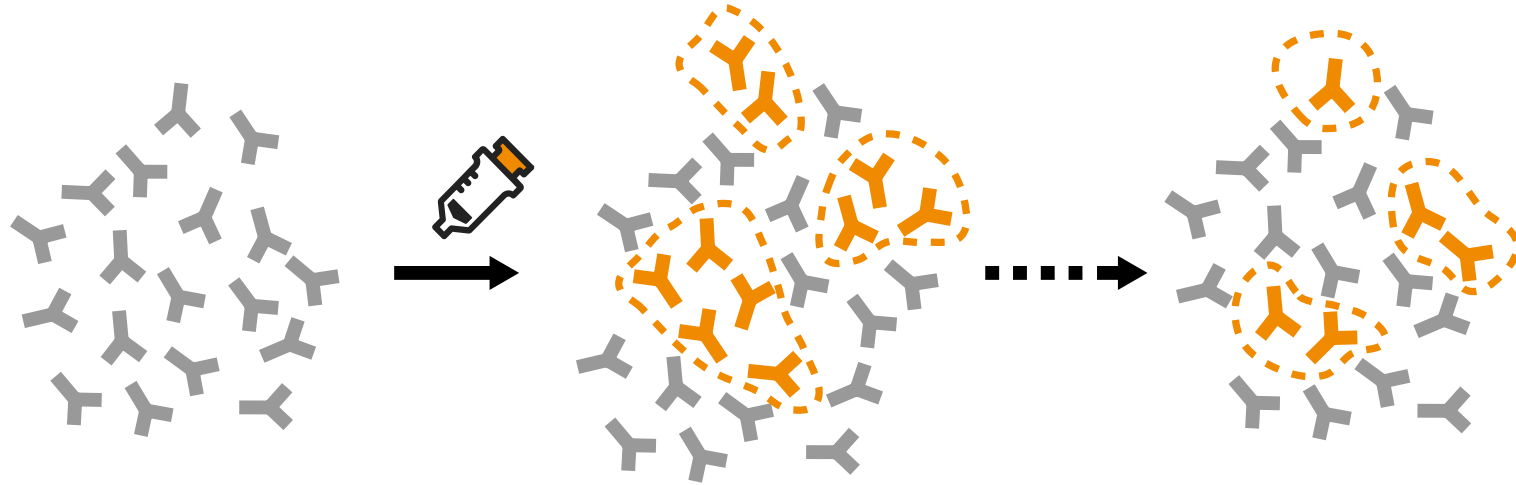
October 25th 2026, Normal and Malignant B-cells workshop



Tracking lineages fates'

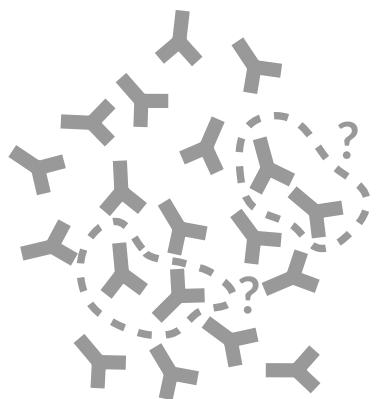
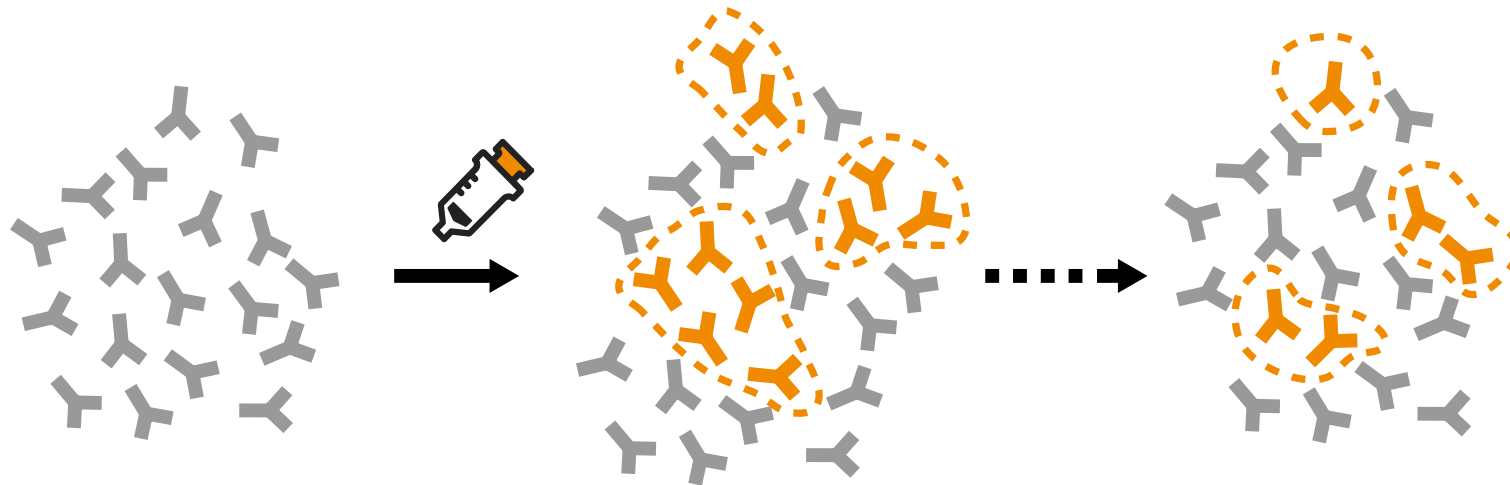


Tracking lineages fates'

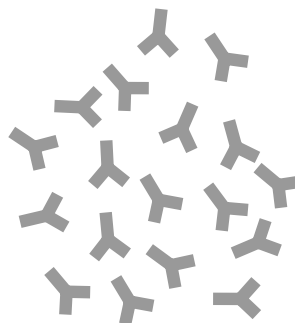


Can we identify
the responding
lineages?

Tracking lineages fates'

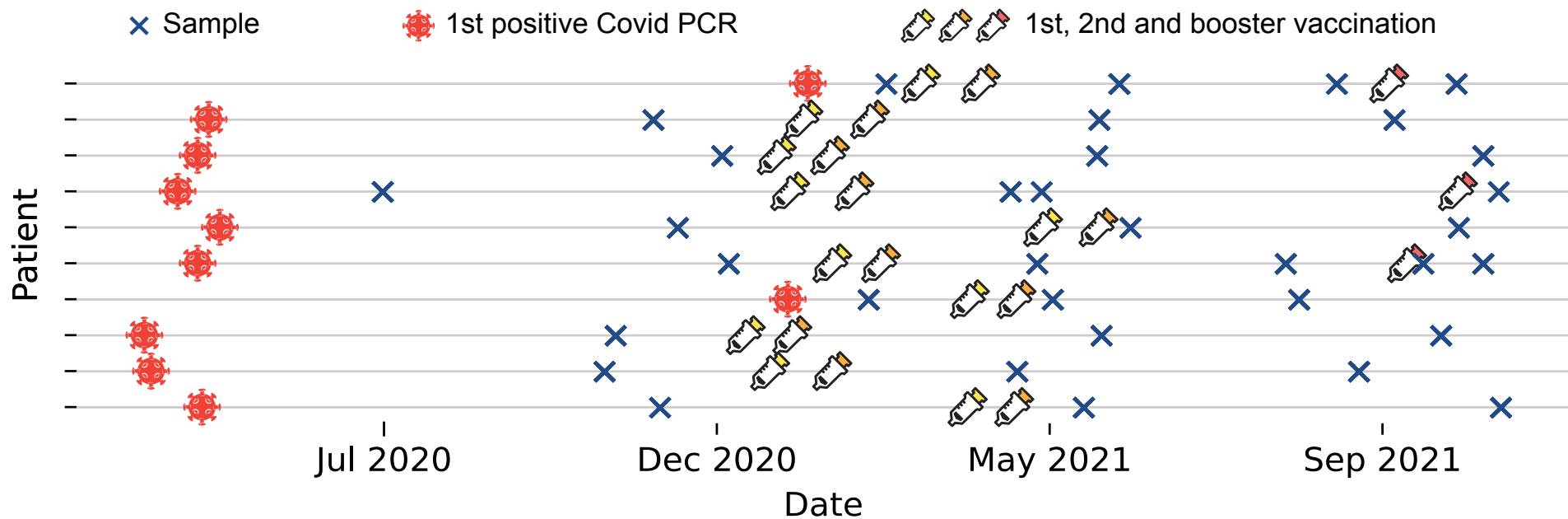


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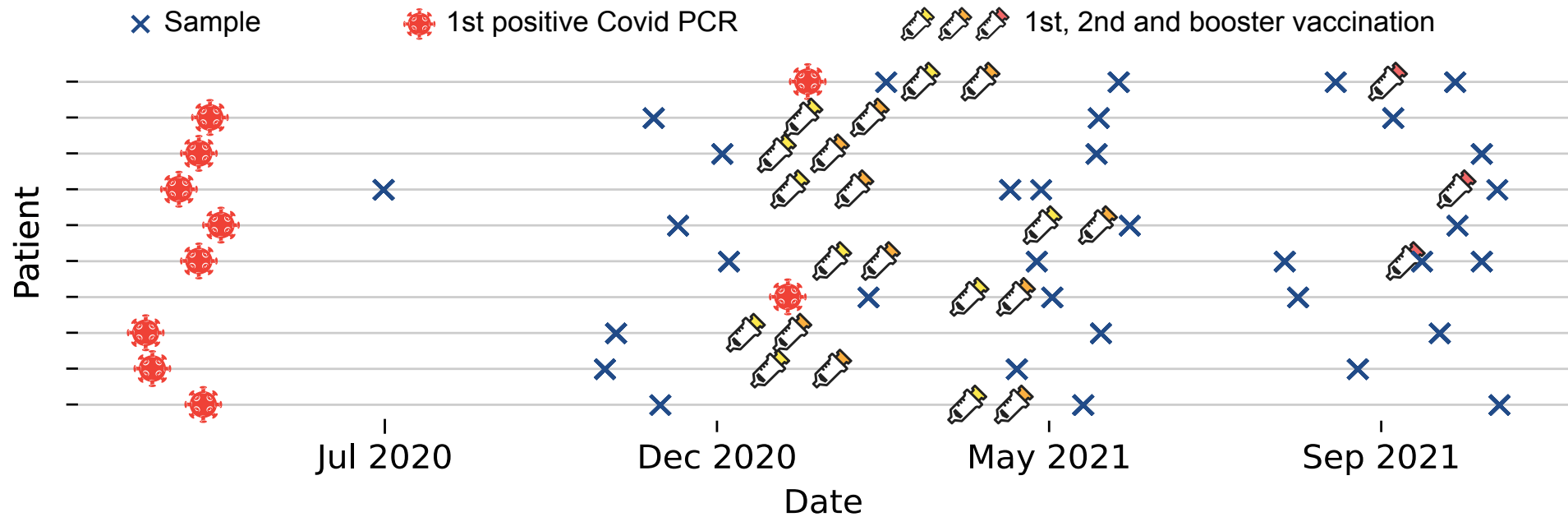
Can we predict
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Longitudinal Repertoire Sequencing



Longitudinal sampling on 10 individuals. Collaboration with the Yu lab at the Ragon institute, 10^9 B cells. Heavy & Light chain.

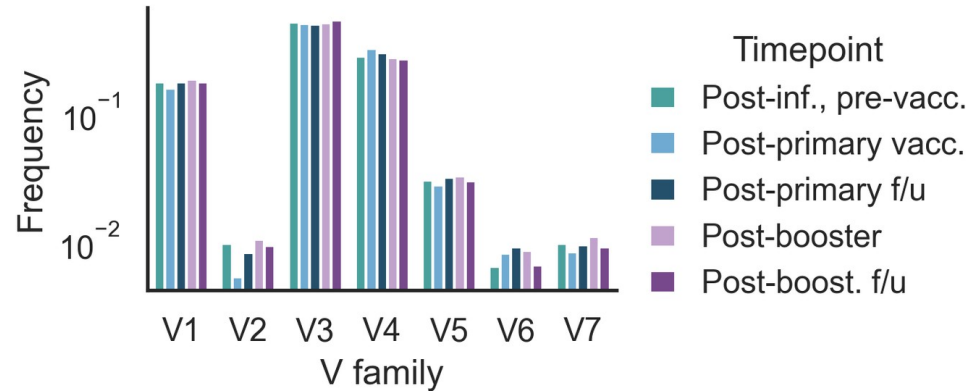
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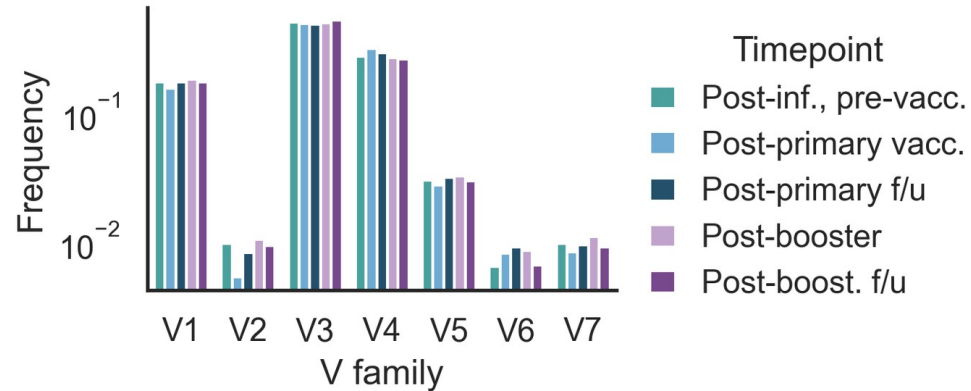
Global statistics don't show the responding lineages

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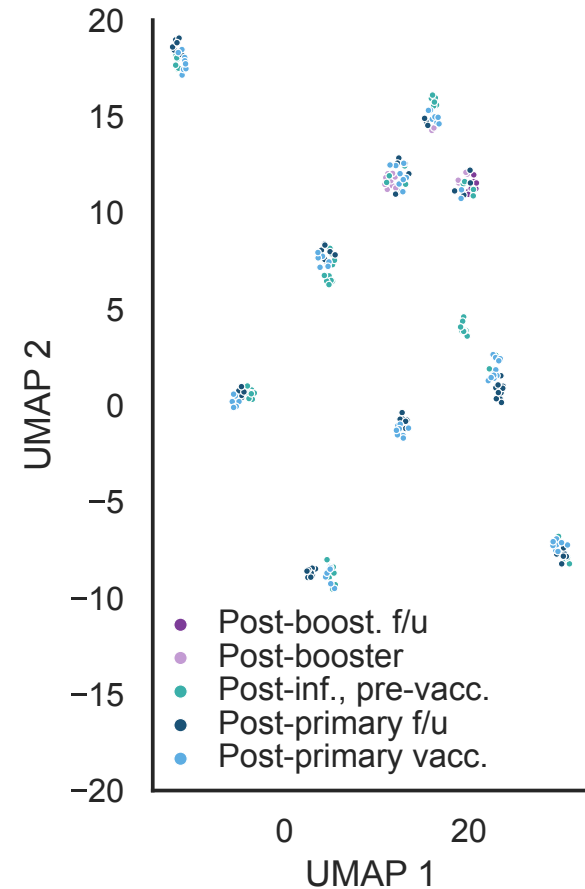


The repertoire is stable over time

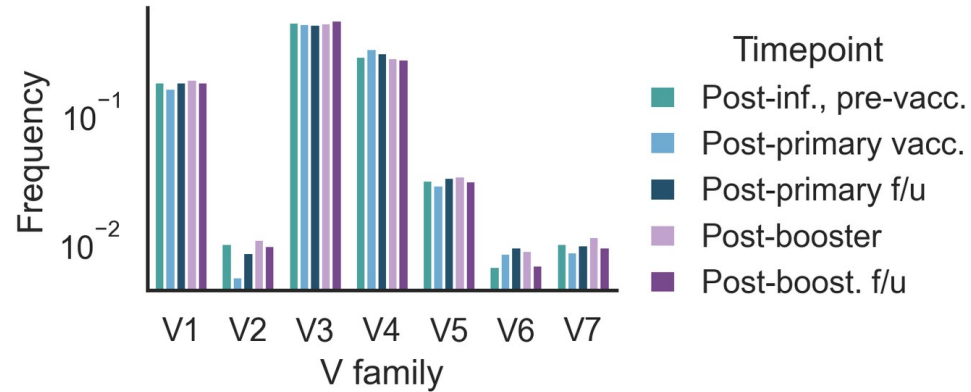
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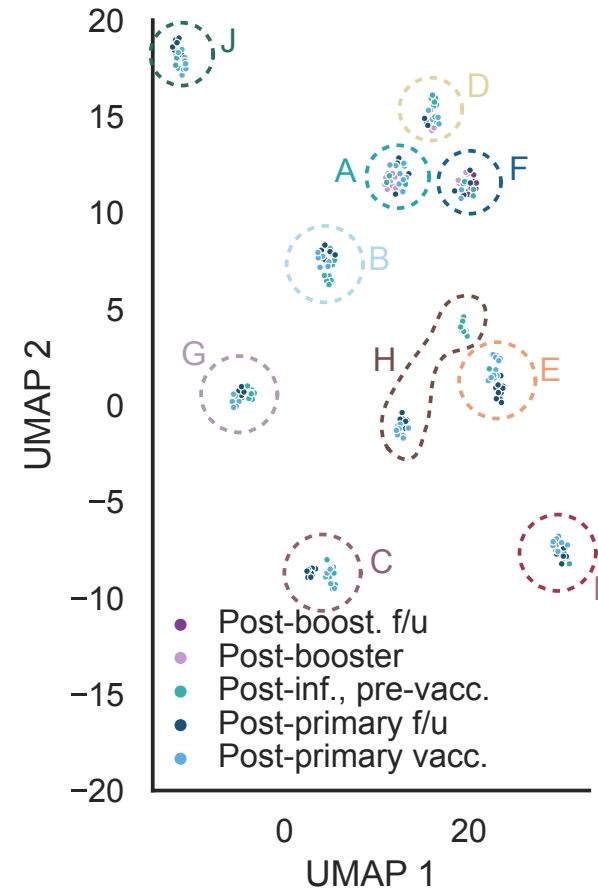
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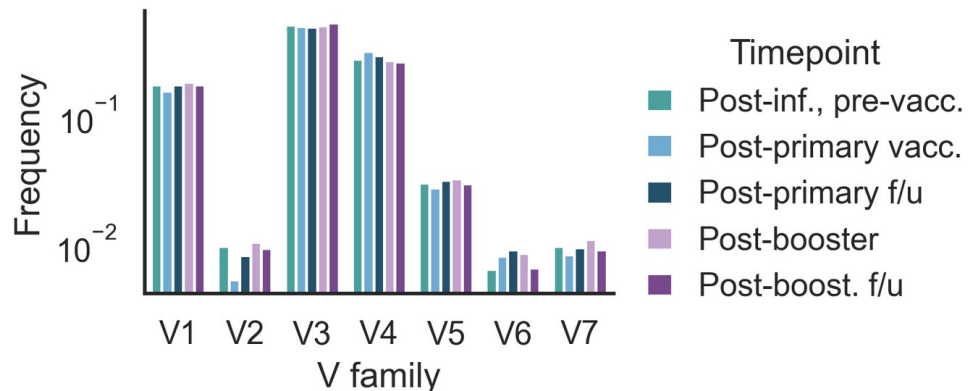
Global statistics don't show the responding lineages



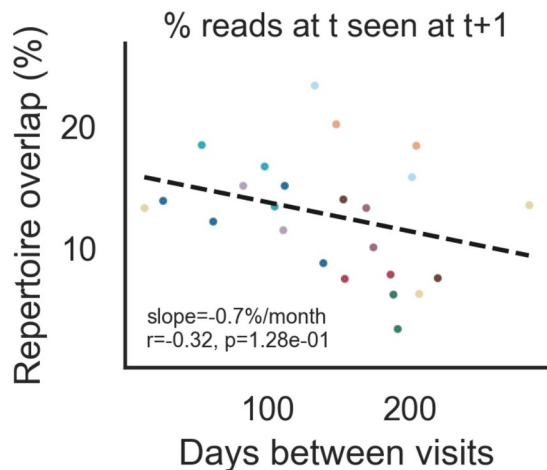
The repertoire is stable over time



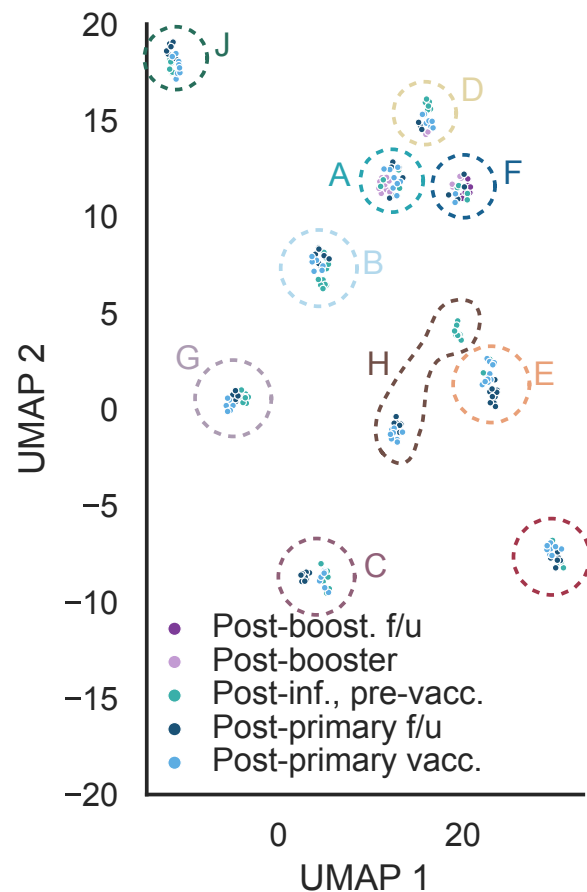
Global statistics don't show the responding lineages



The repertoire is stable over time

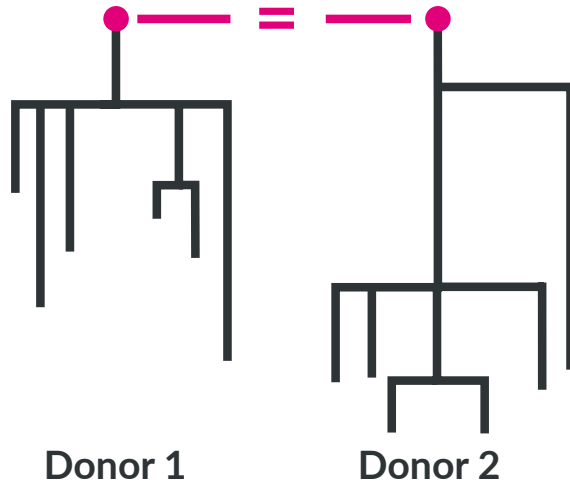


This stability is maintained through turnover



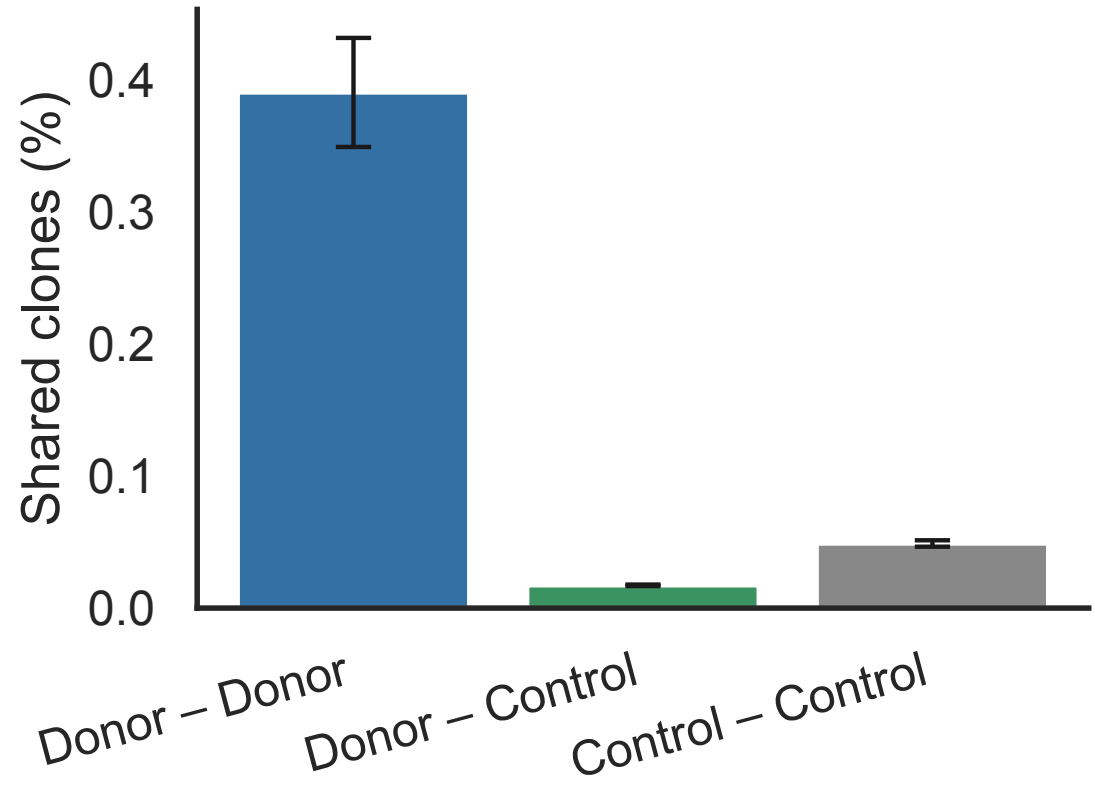
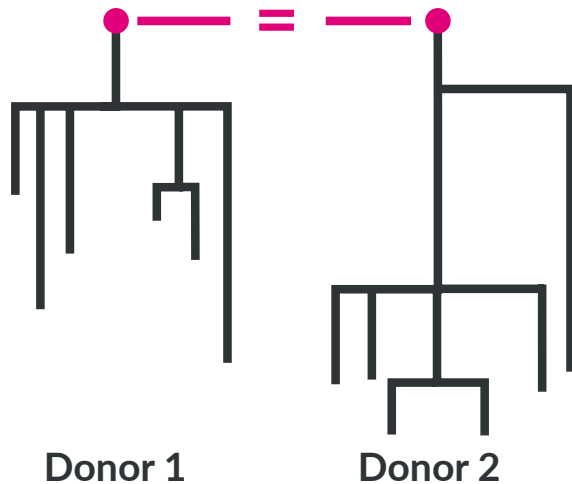
Shared lineages as a proxy for specificity ?

Shared lineages =
same unmutated
ancestor



Shared lineages as a proxy for specificity ?

Shared lineages =
same unmutated
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Shared lineages are enriched for response signature

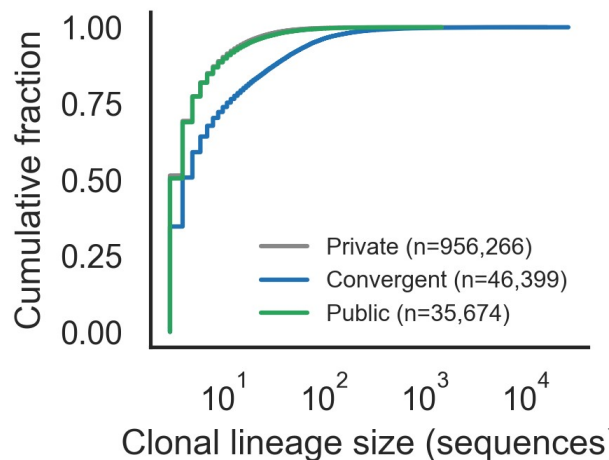
Public clones = shared with the healthy controls

Convergent clones = with the cohort

Shared lineages are enriched for response signature

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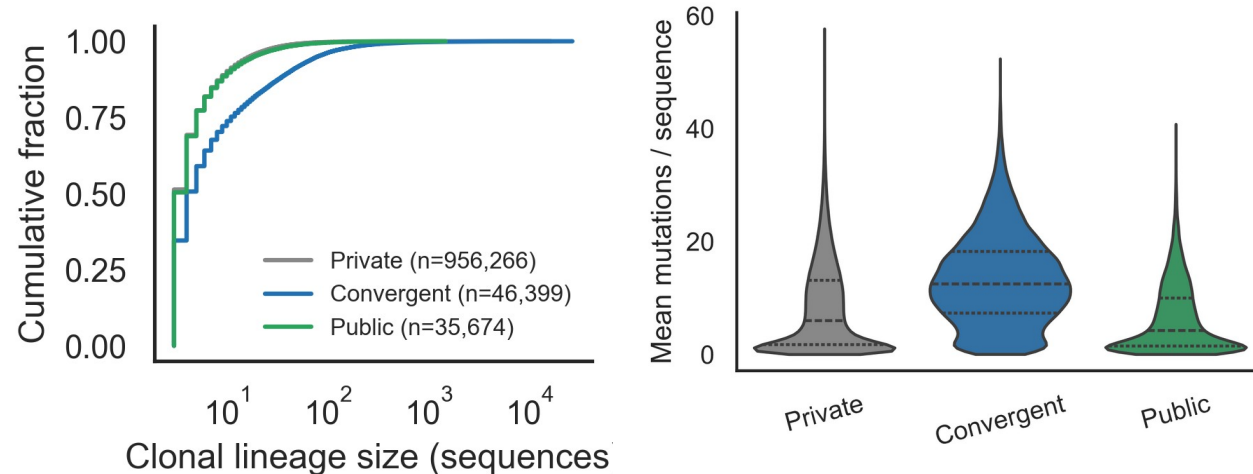
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Shared lineages are enriched for response signature

Public clones = shared with the healthy controls

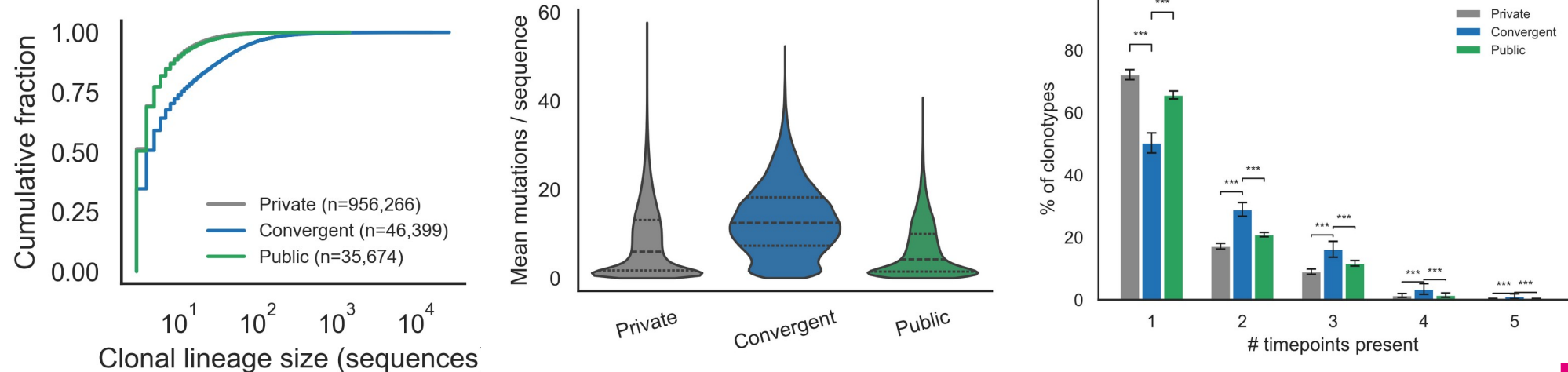
Convergent clones = with the cohort



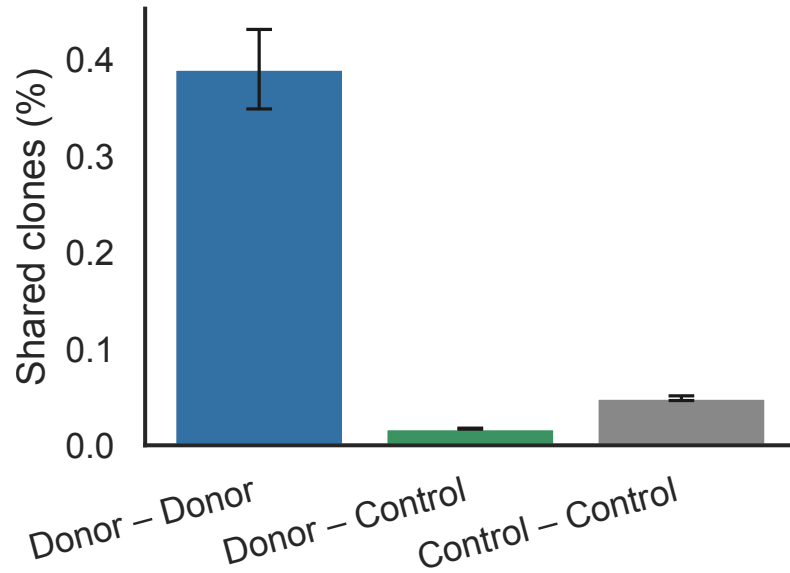
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Public clones = shared with the healthy controls

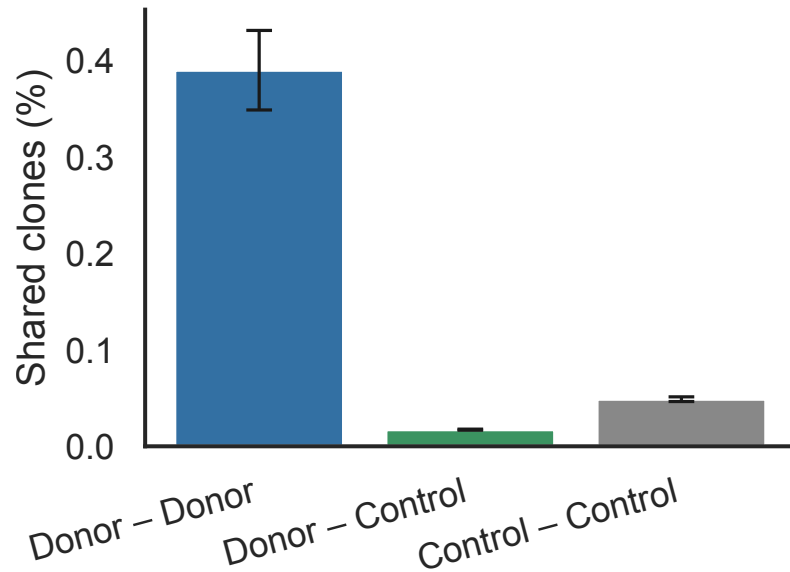
Convergent clones = with the cohort



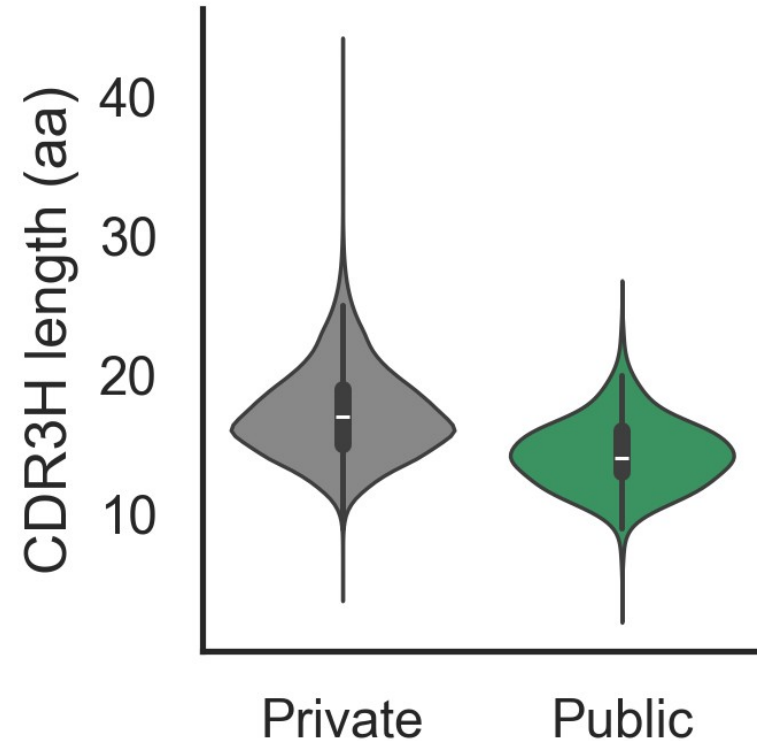
Shared lineages are a biased sample



Shared lineages are a biased sample



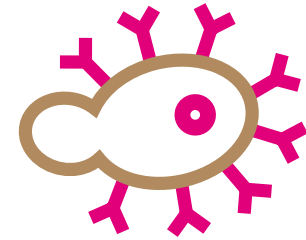
Shared BCR are shorter,
and more often generated
by recombination.



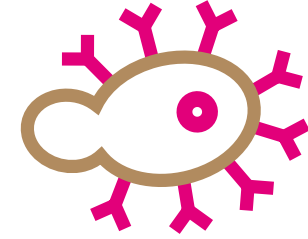
Measuring specificity

Measuring specificity

In-vitro “clone” of the repertoire.



Measuring specificity

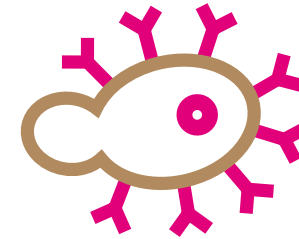


In-vitro “clone” of the repertoire.

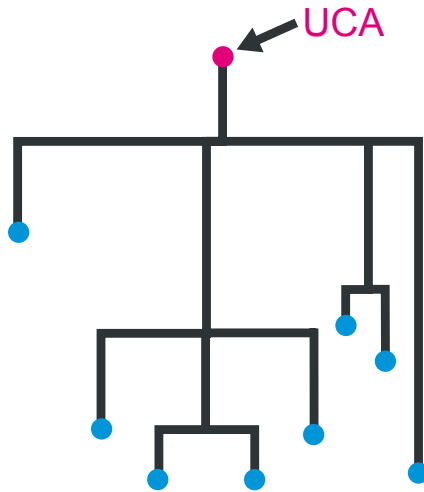
⇒ Quantitative readout

⇒ Control over the antibodies tested

Measuring specificity



In-vitro “clone” of the repertoire.

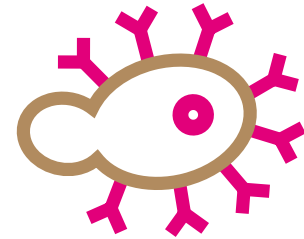


⇒ Quantitative readout

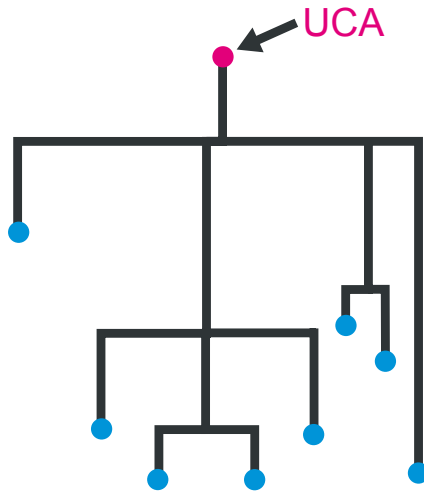
⇒ Control over the antibodies tested

We recreated the **inferred ancestor (UCA)** for the largest lineages.

Measuring specificity



In-vitro “clone” of the repertoire.



⇒ Quantitative readout

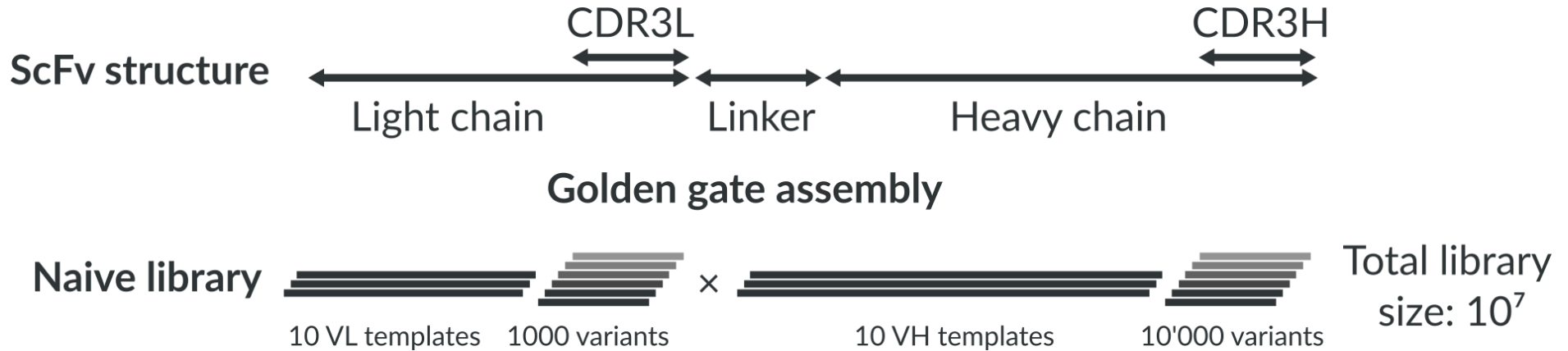
⇒ Control over the antibodies tested

We recreated the **inferred ancestor (UCA)** for the largest lineages.

Bulk sequencing ⇒ No heavy-light pairing
⇒ Combinatorial approach



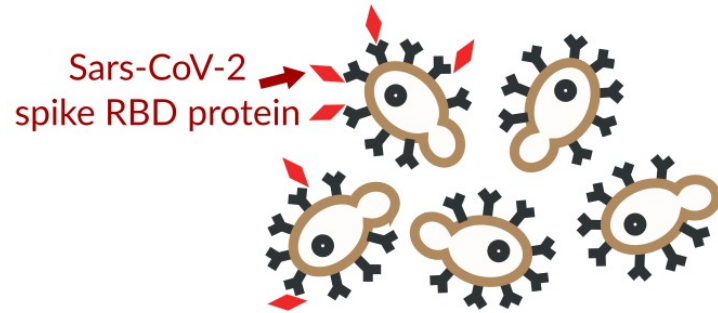
Making the library



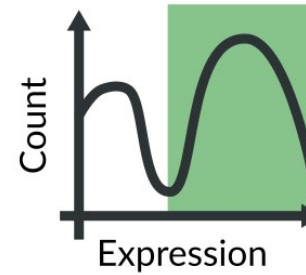
We focus on the **10'000 largest clonal lineages.**

(still a very small part of the overall repertoire ~ 0.1%)

Measuring Affinities in High-Throughput

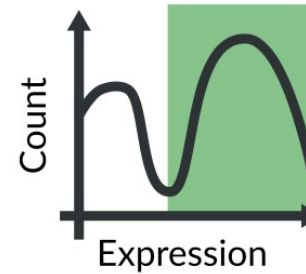
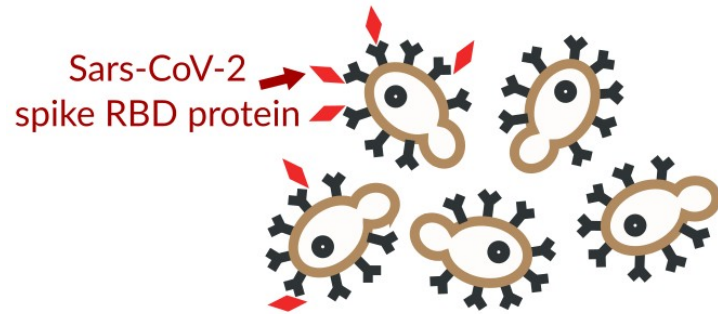


Yeast expression



Sort for
expression
and **binder**
enrichment

Measuring Affinities in High-Throughput



Sort for
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and **binder**
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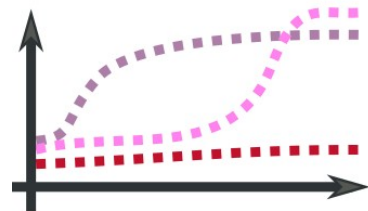


Barcode-scFv association



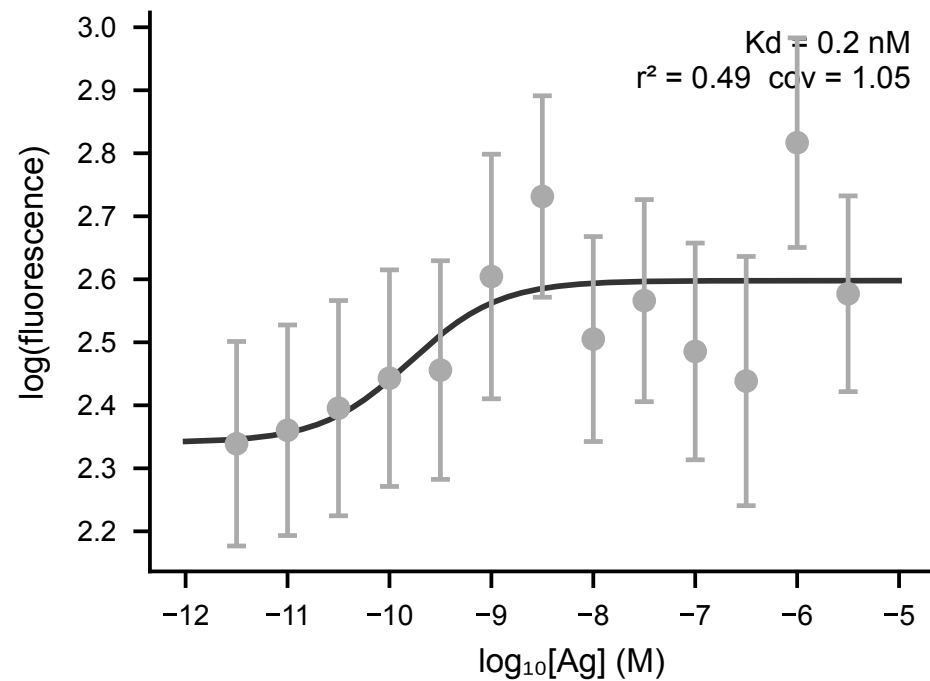
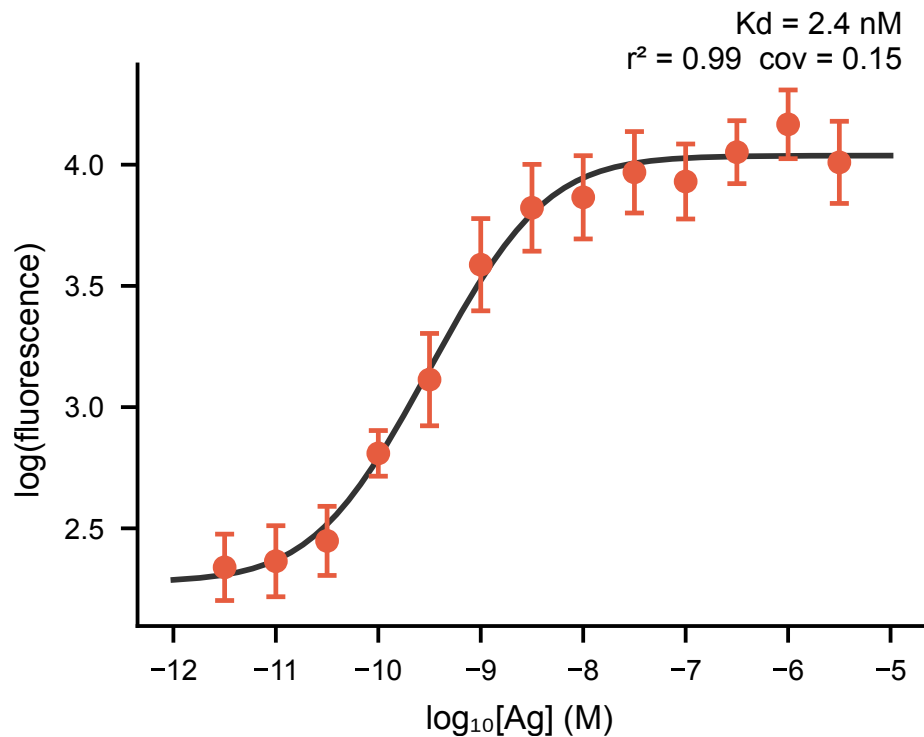
+ Nanopore sequencing

Library size = 68'000

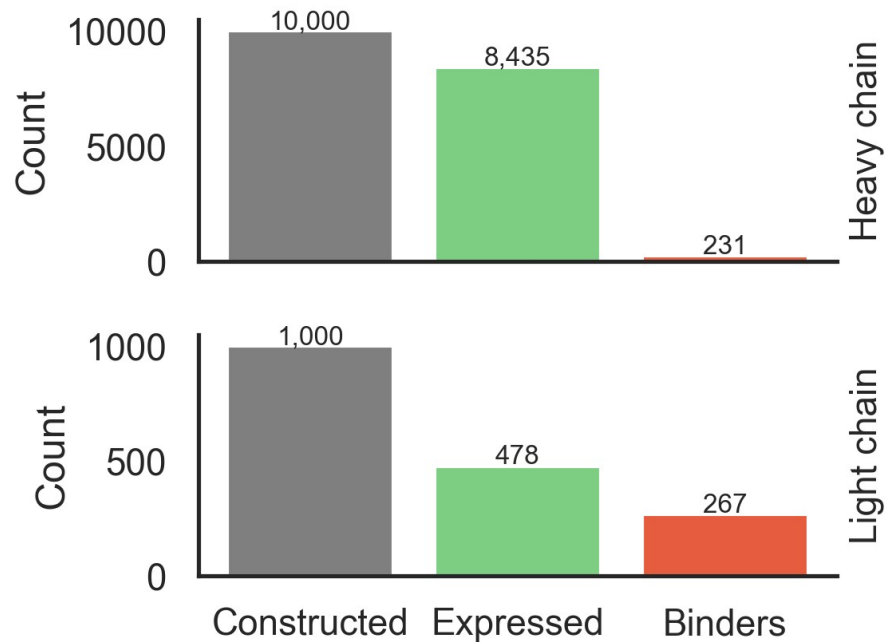


Library size = 10'000

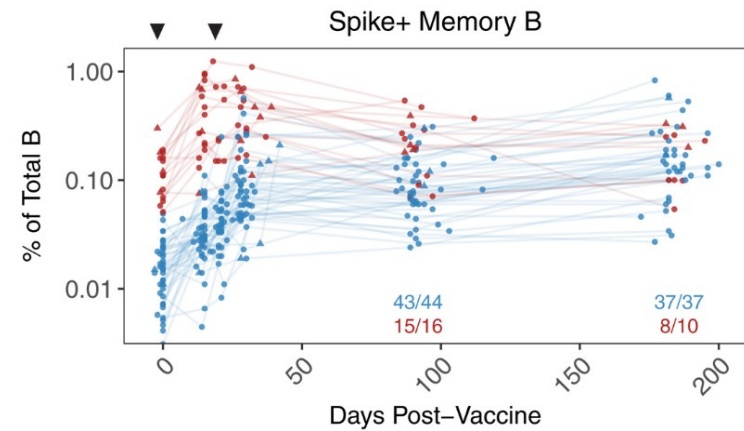
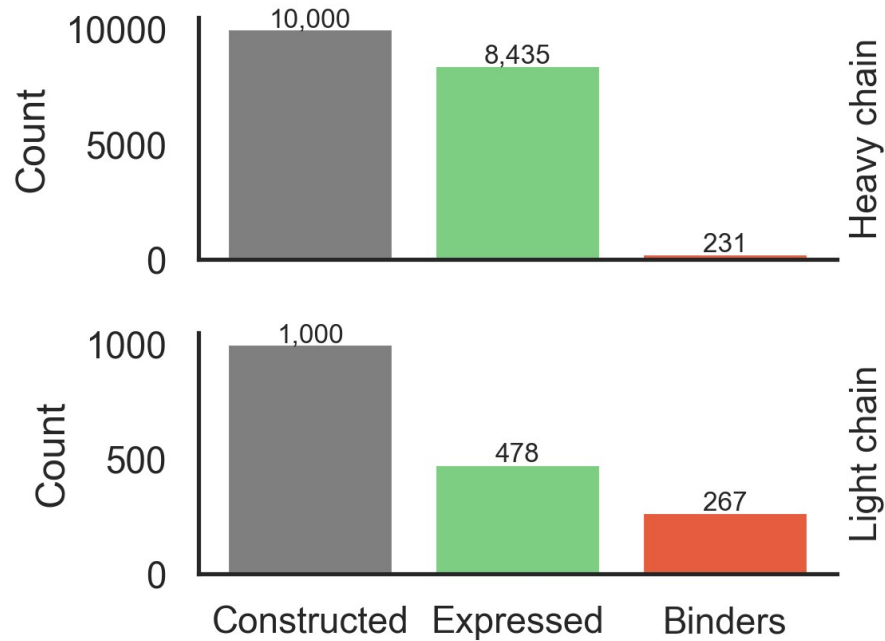
Measuring Affinities in High-Throughput



2% of these lineages bind the RBD

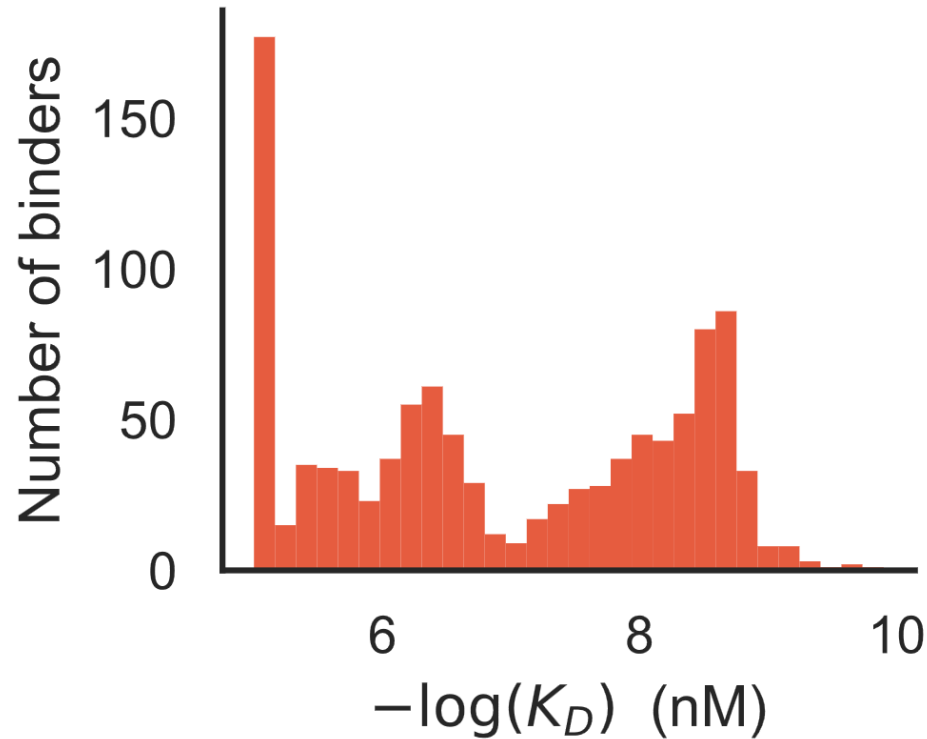


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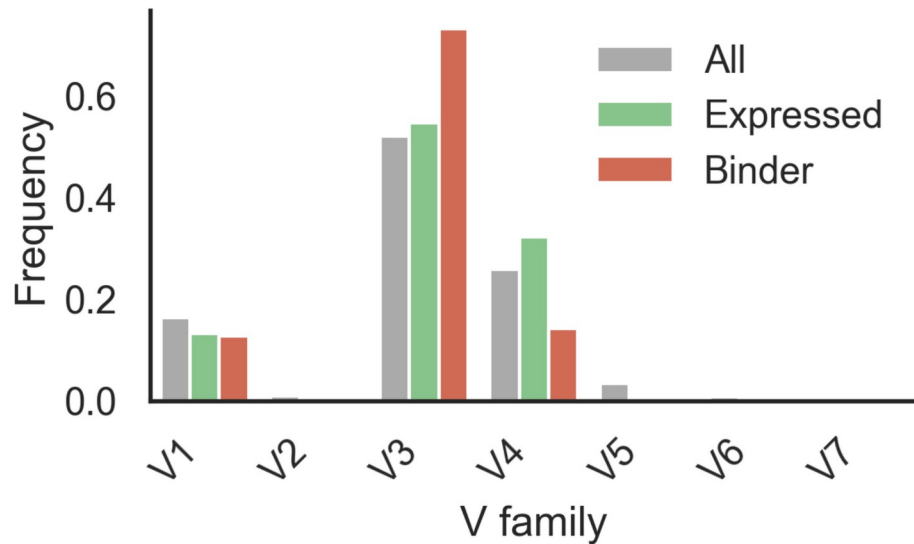
Goel, Rishi R. et al. Science (2021)

Unmutated ancestors span a wide affinity range



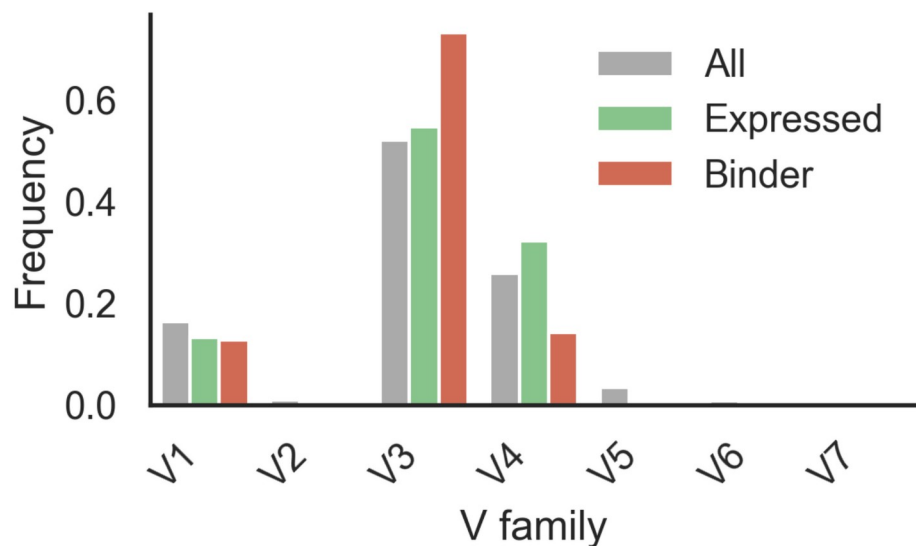
Binders come from across the repertoire

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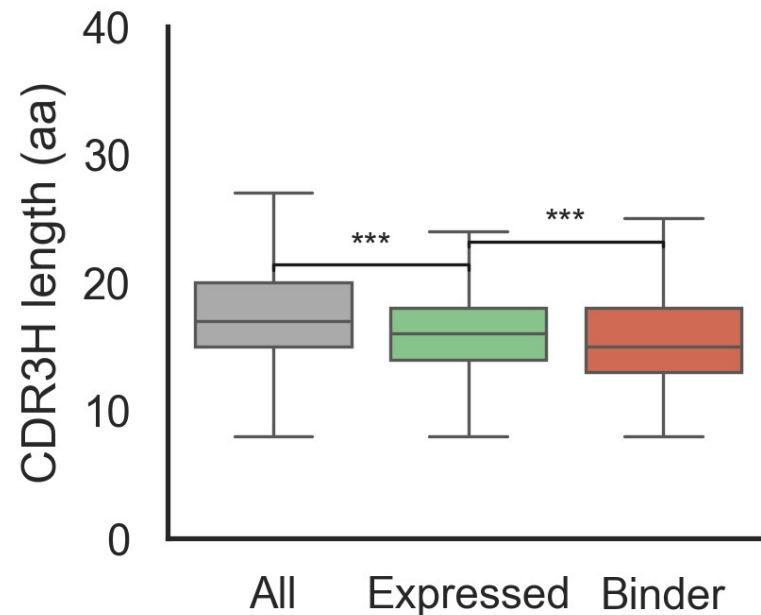


Slightly skewed V-gene
distribution

Binders come from across the repertoire



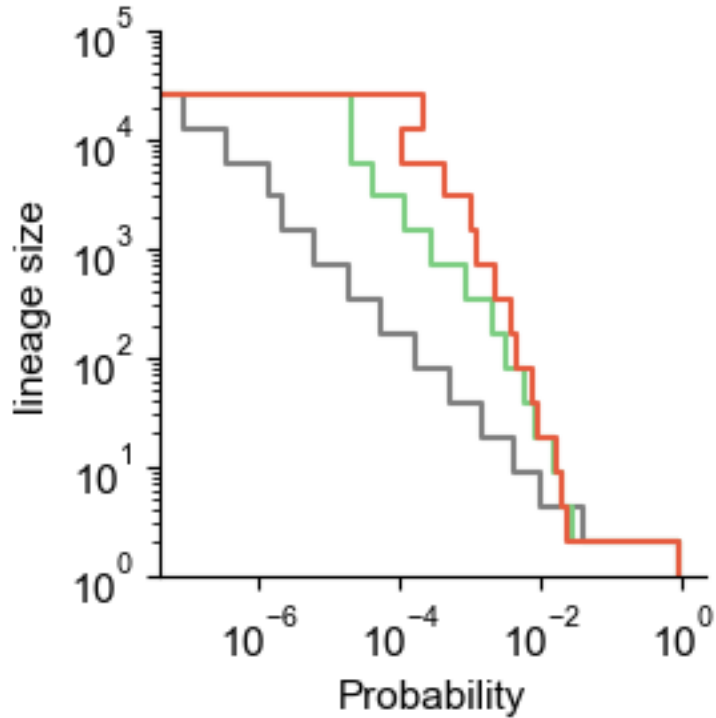
Slightly skewed V-gene distribution



Binders tend to be shorter

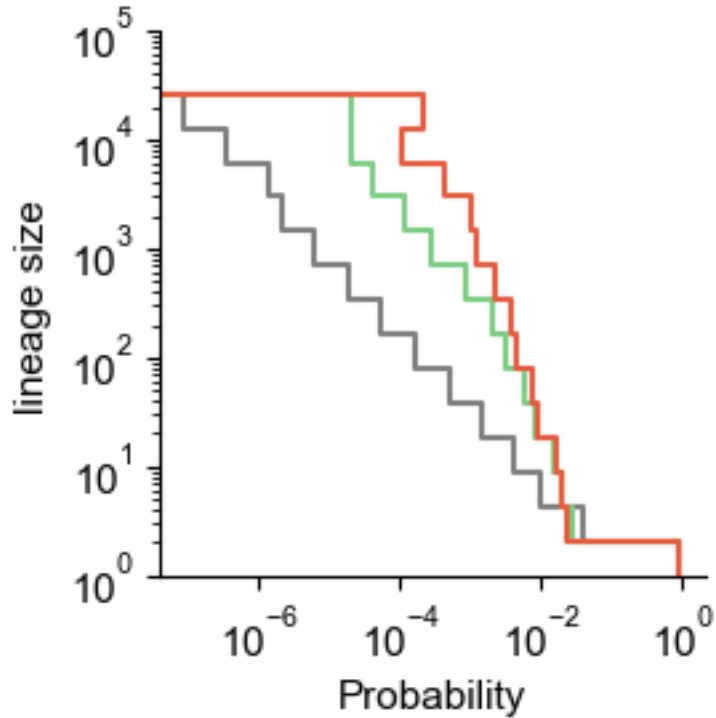
Binder lineages have a response signature

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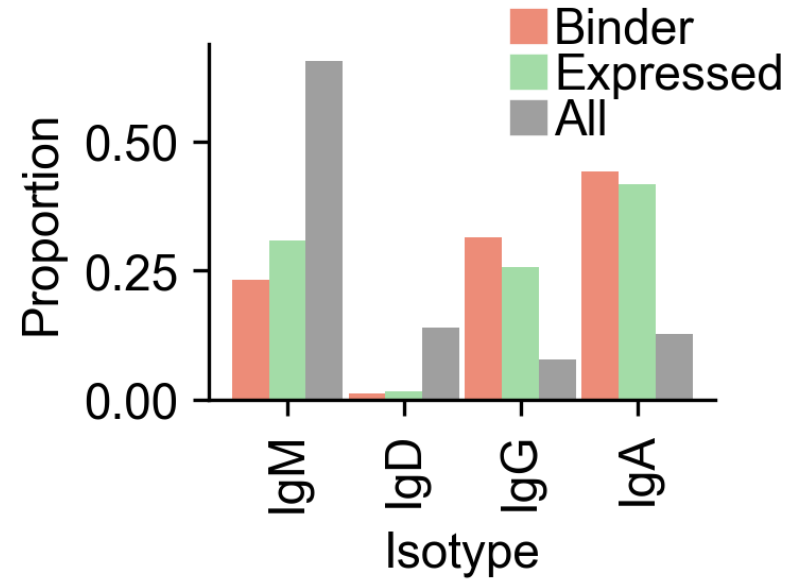


Binder lineages are larger

Binder lineages have a response signature

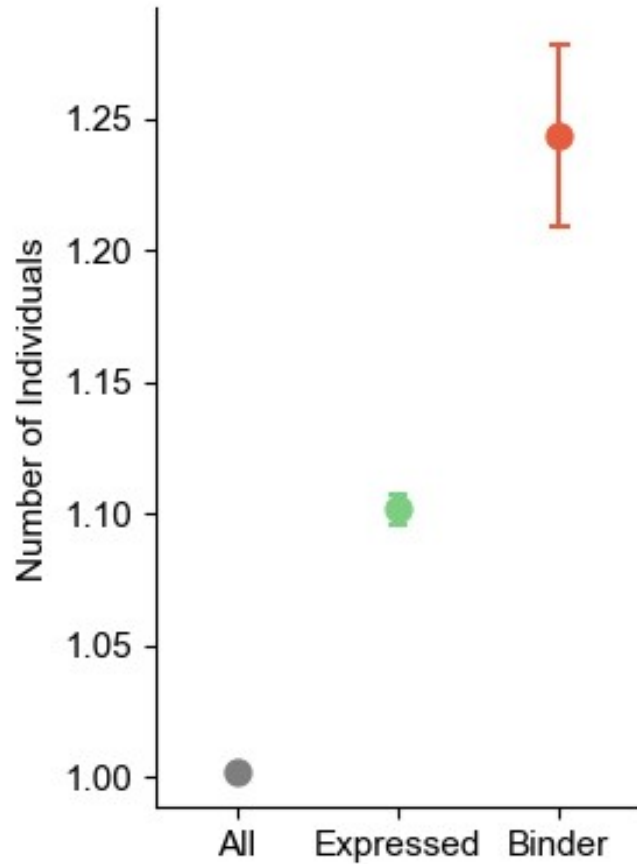


Binder lineages are larger

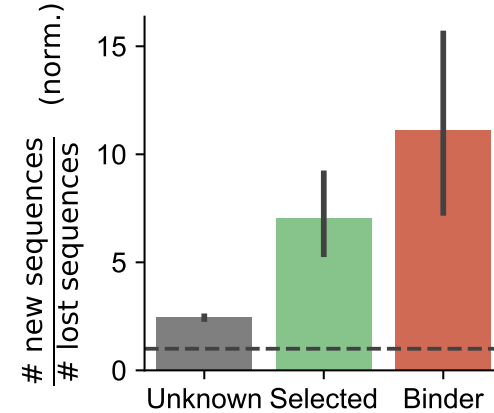
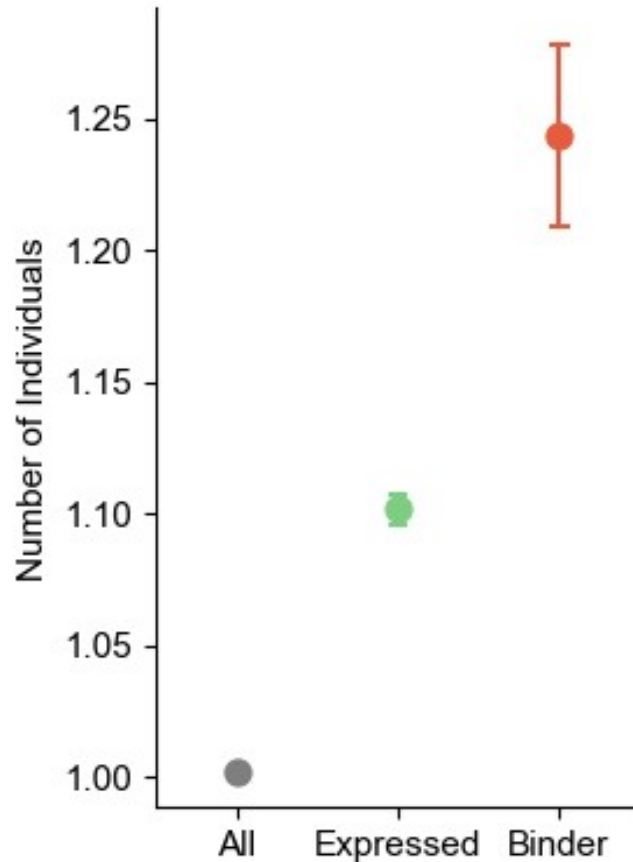


Binder lineages are more often class-switched

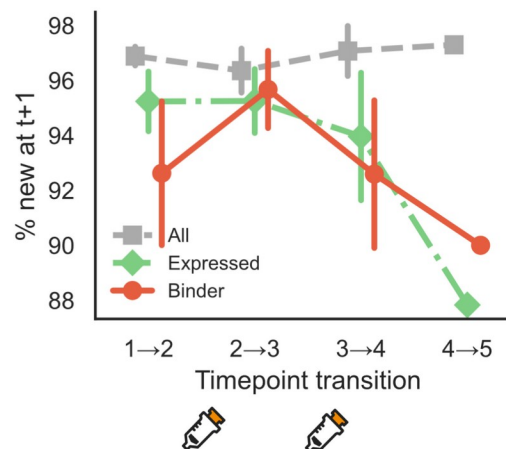
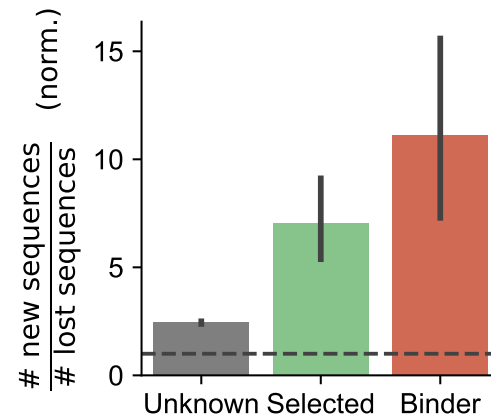
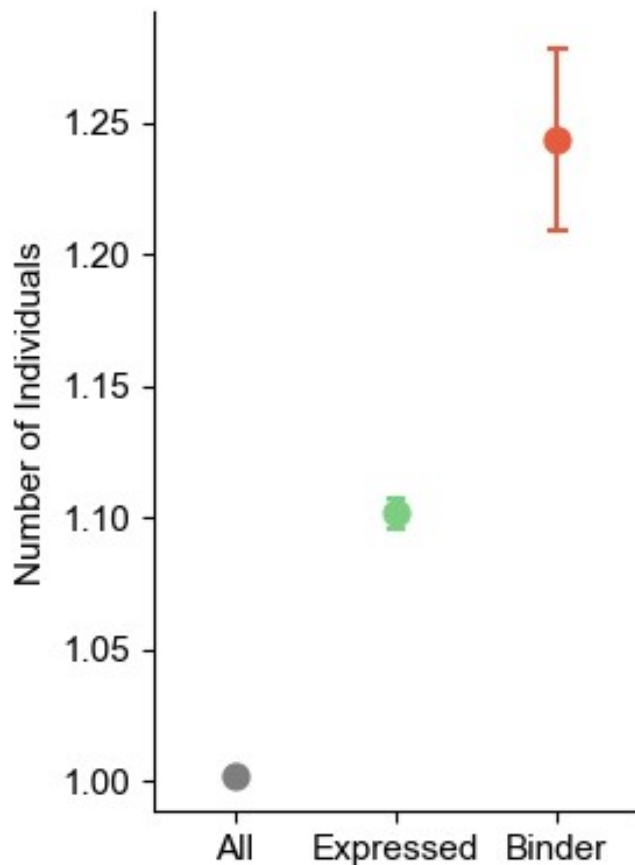
Binder lineages are shared and persistent



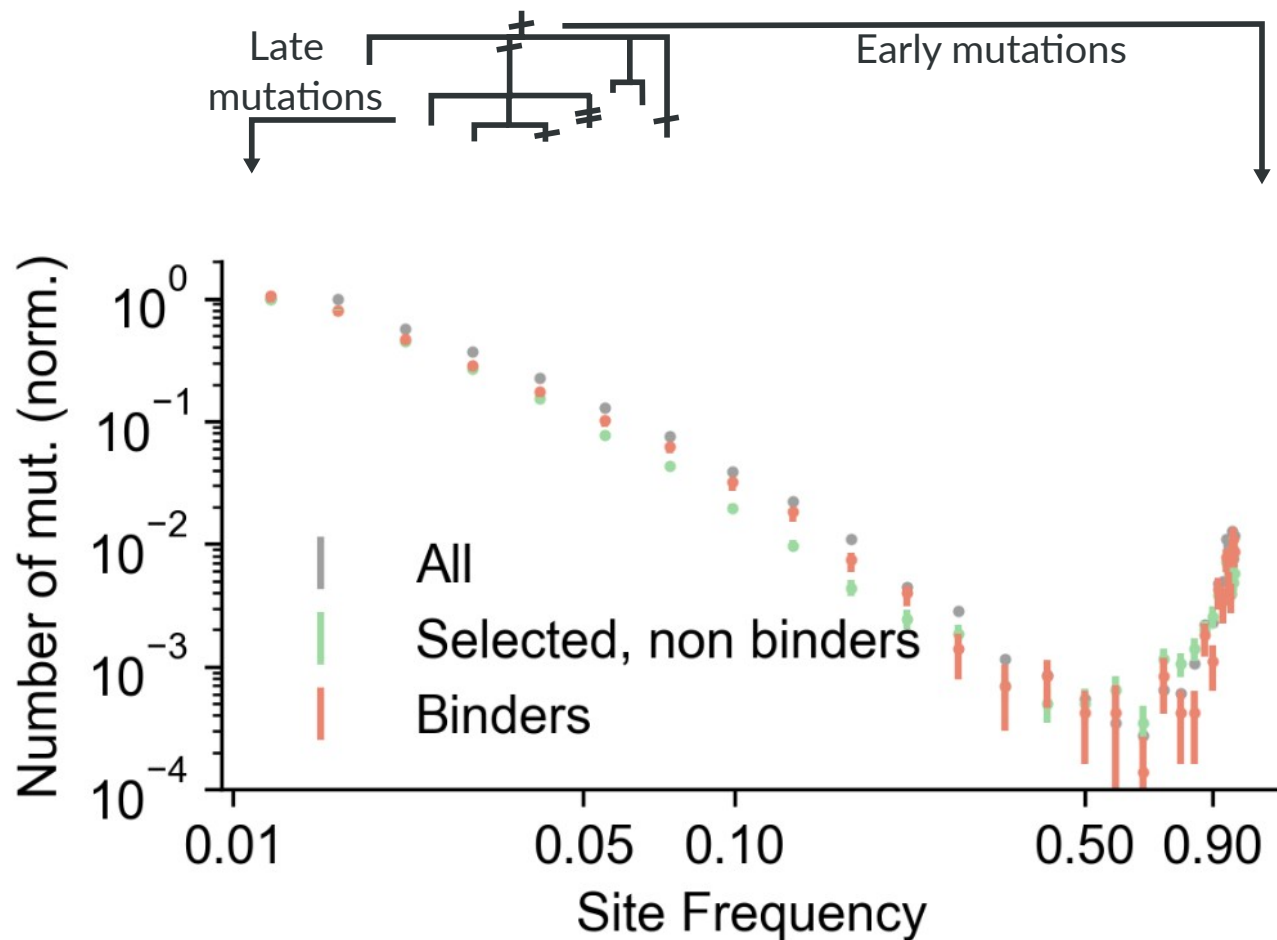
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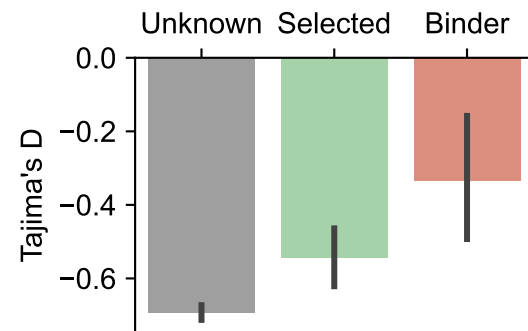
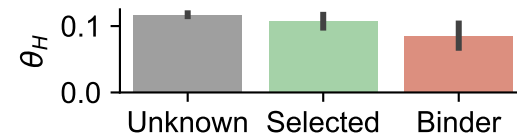
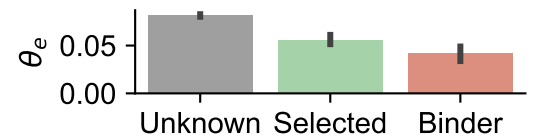
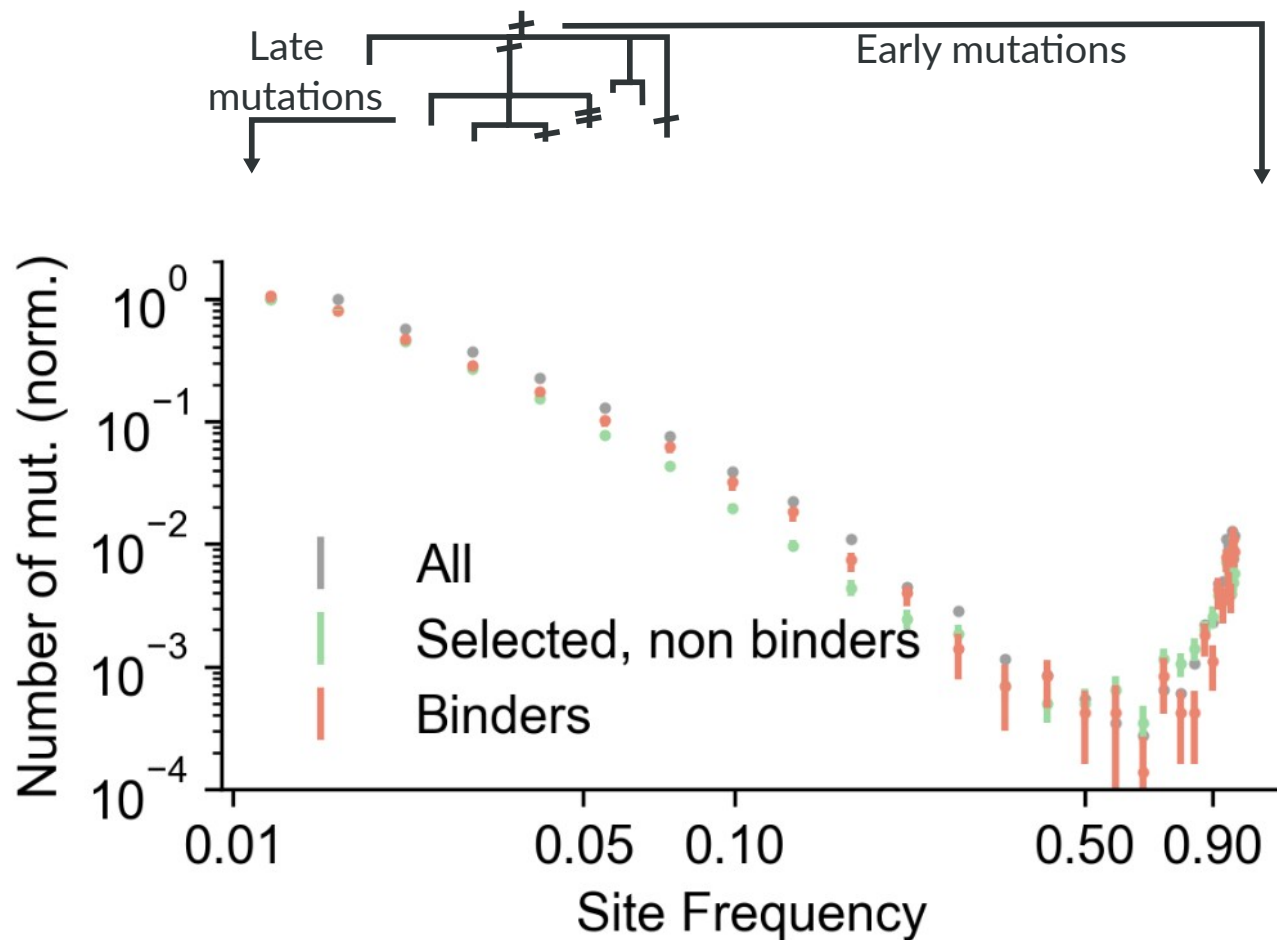
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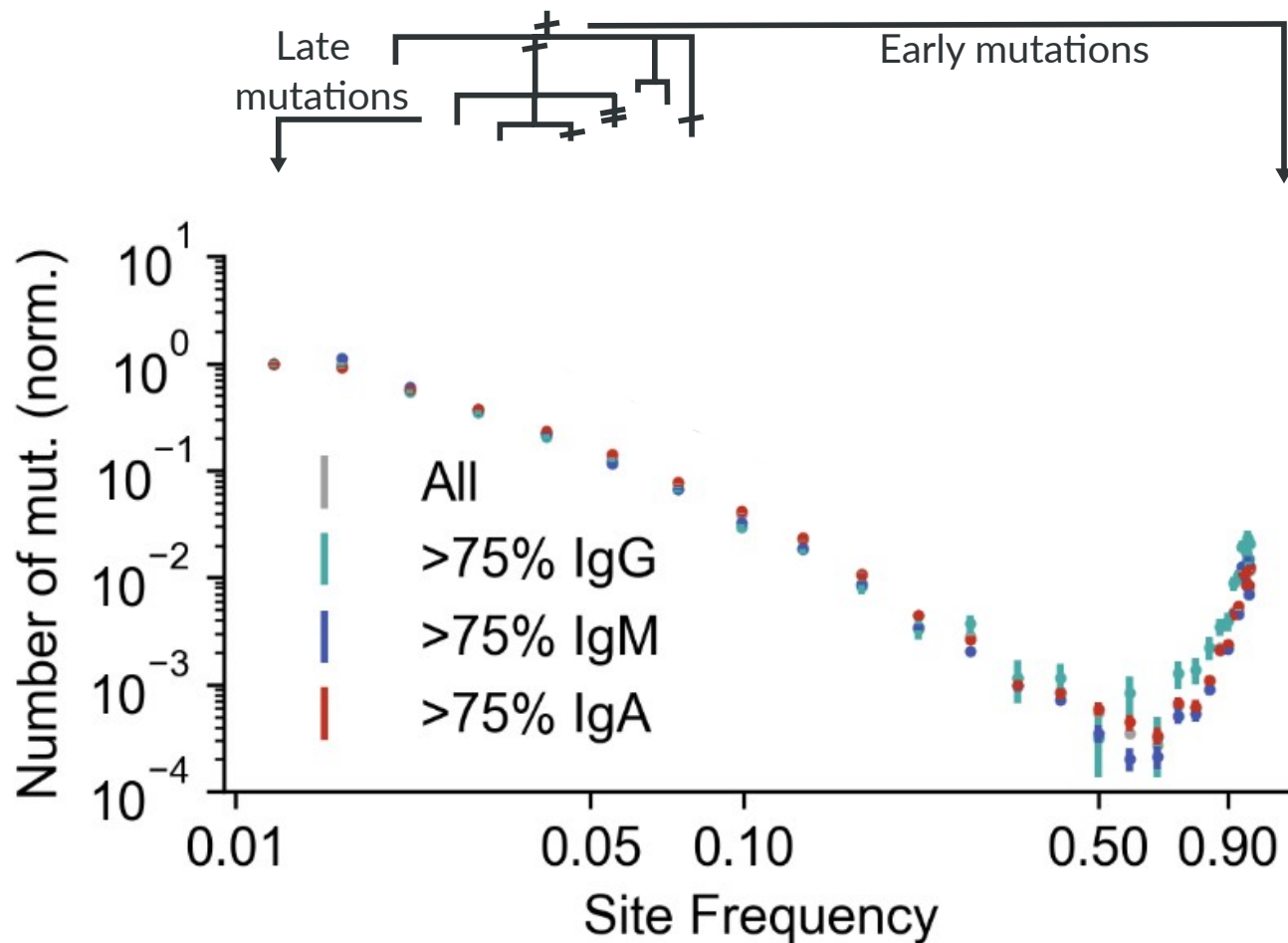
Binder lineages are under stronger selection



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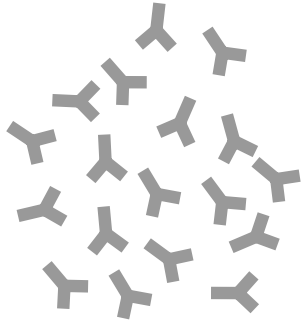


Can we identify the responding lineages ?



Binder lineages have distinct properties, but none of them is discriminative enough to identify binders from sequence alone.

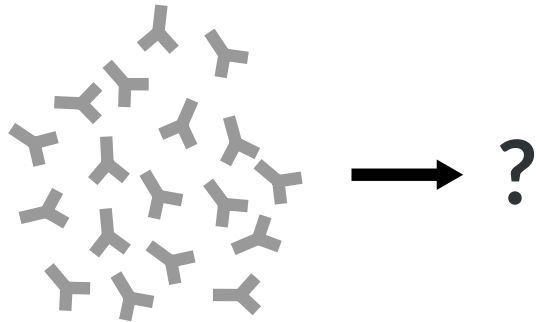
Shared lineages as replicate experiments



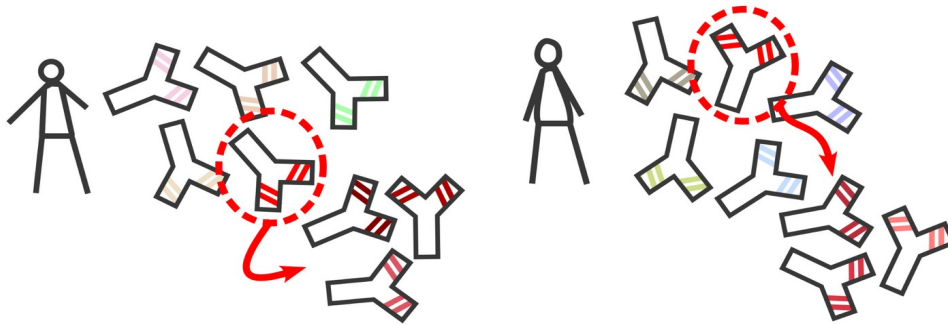
?

Can we predict
the evolution
of a repertoire?

Shared lineages as replicate experiments

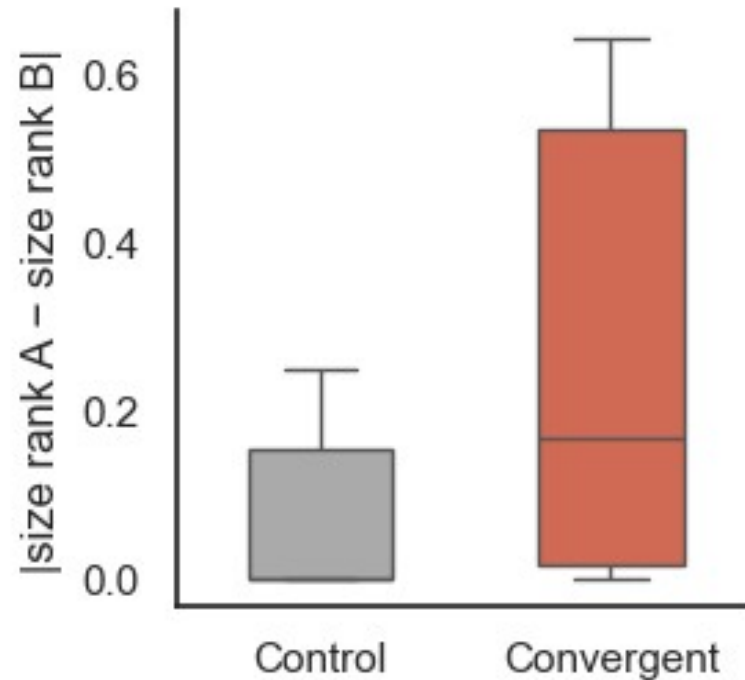
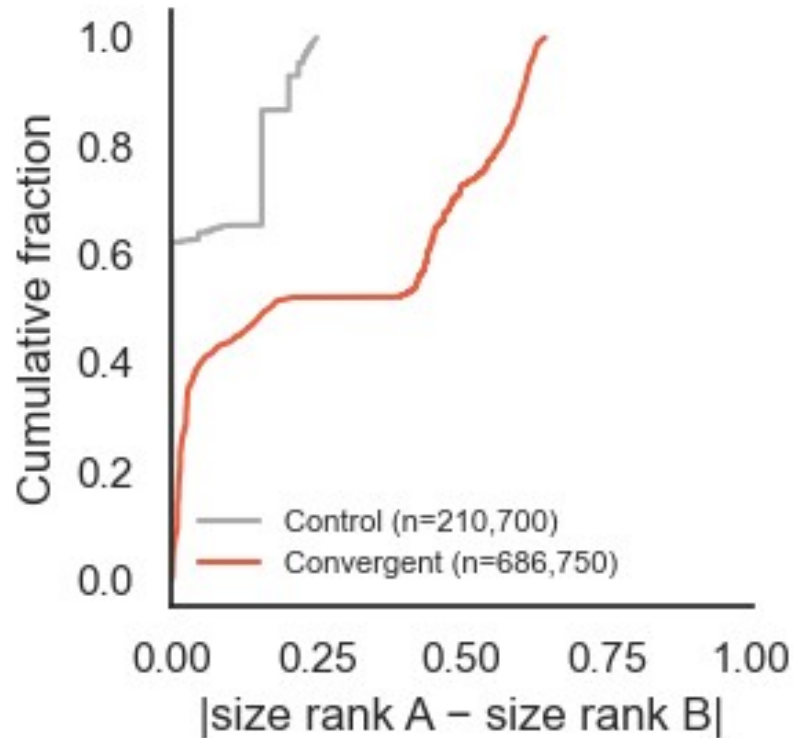


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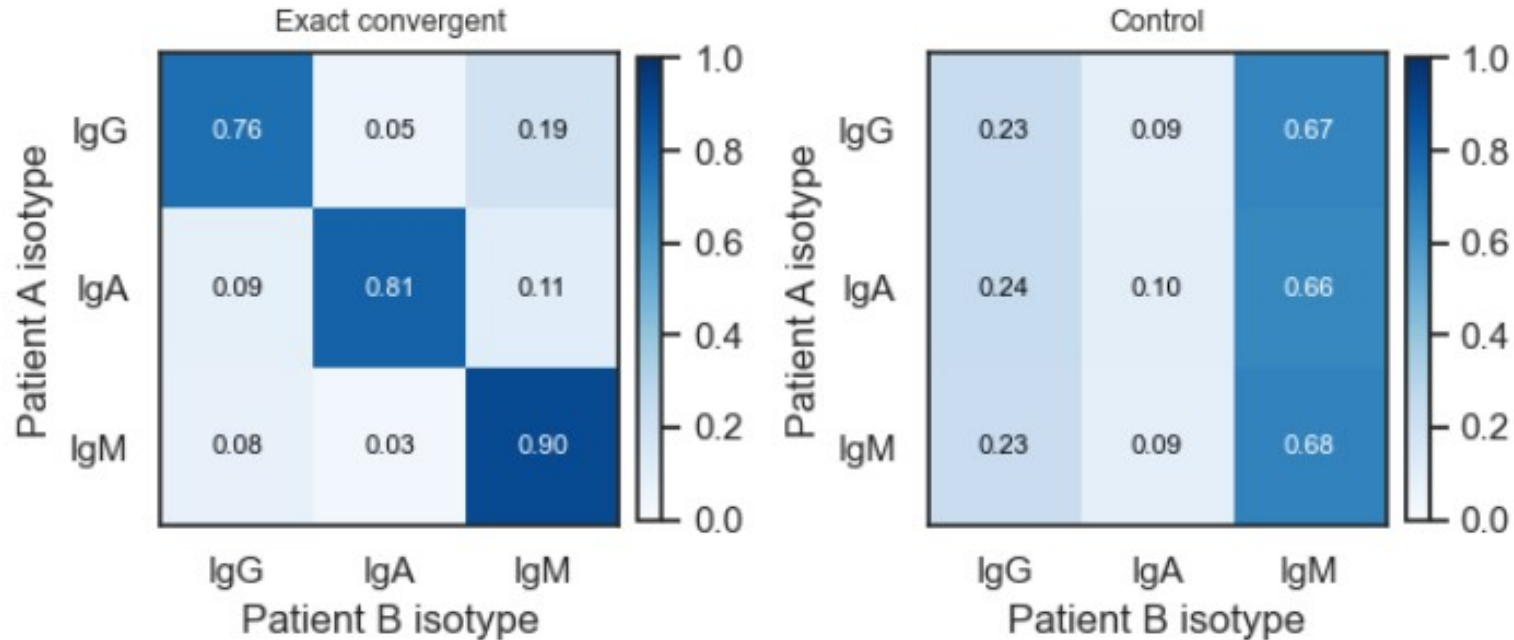


Will lineages with
the same UCA react
in the same way in
different patients?

Shared lineages have similar trajectories

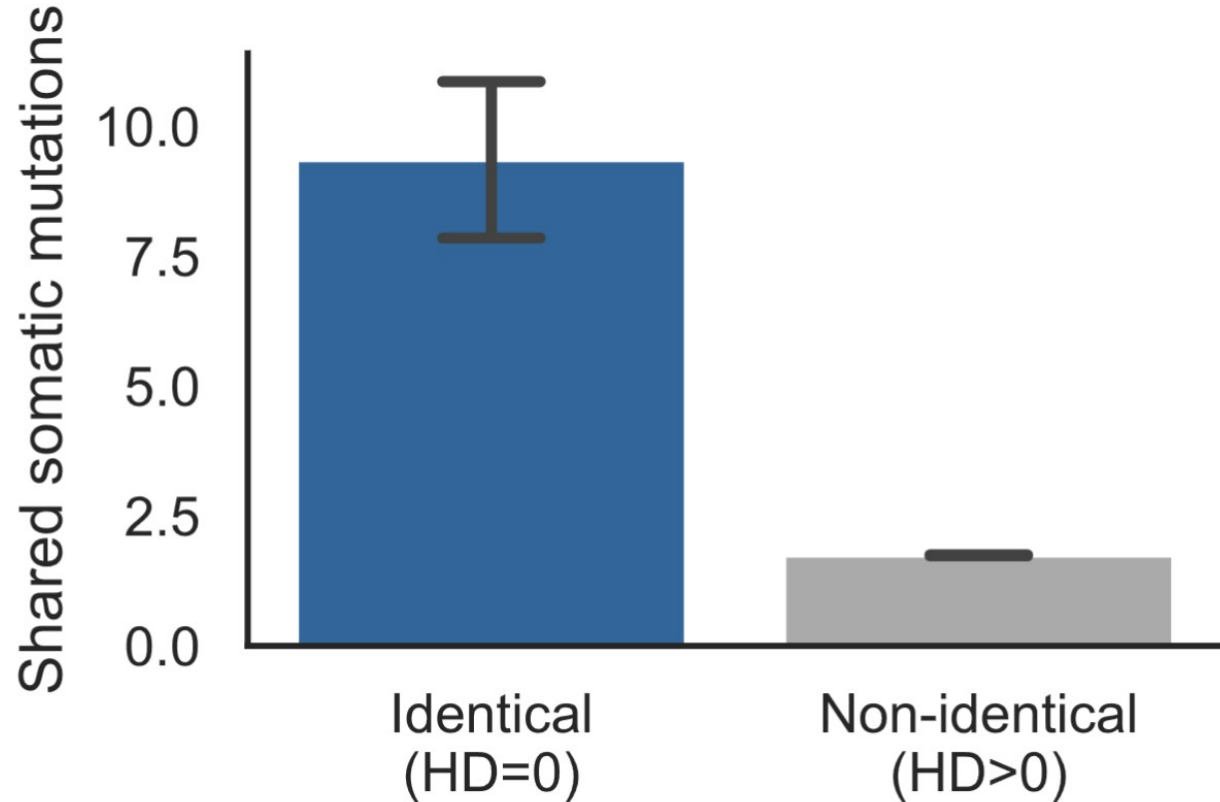


Shared lineages tend to class-switch similarly



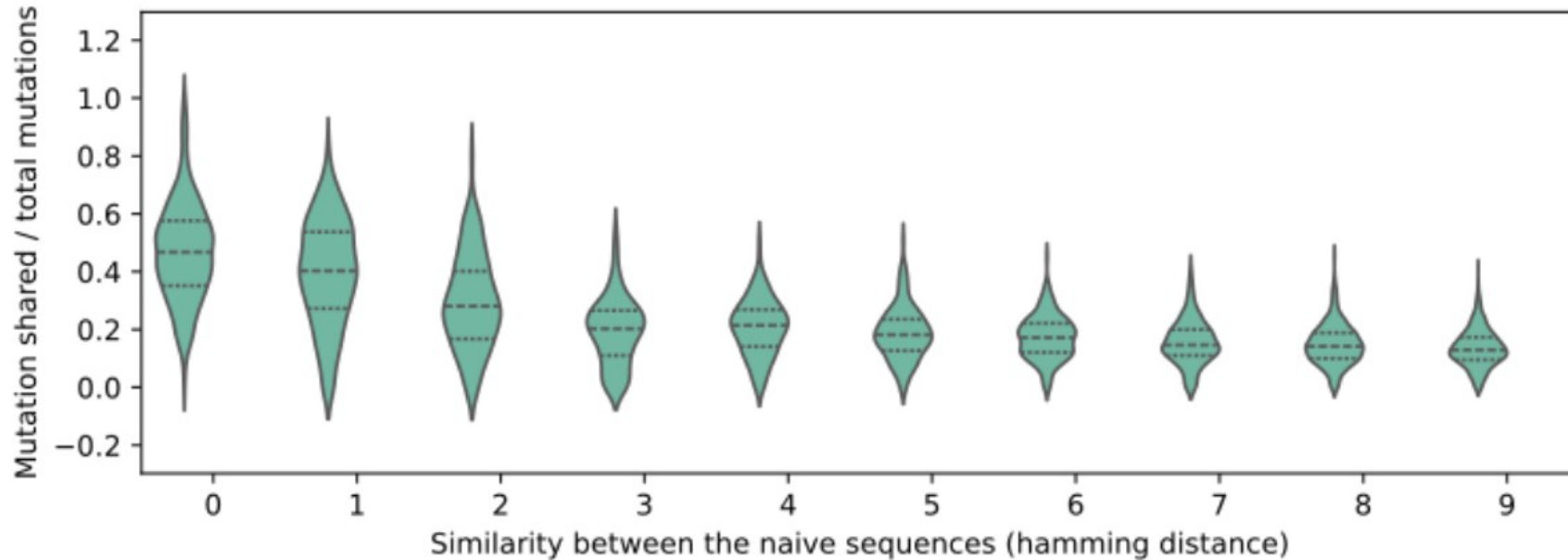
B-cell evolution is reproducible between individuals

Lineages with the same unmutated ancestor accumulate the same mutations



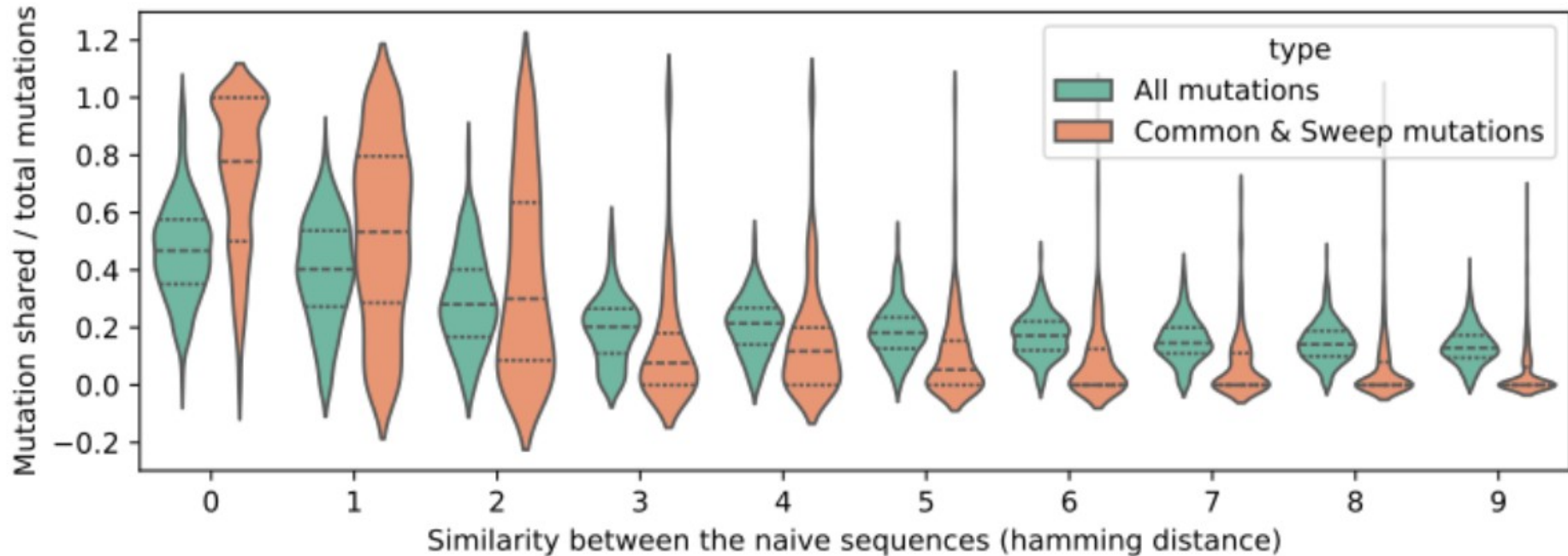
B-cell evolution is reproducible between individuals

But this is less and less true as the distances between the UCAs increase



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Summary

Vaccination experiment. 10^9 B-cells receptors sequenced,
 10^4 antibody-antigen interaction measured

Vaccination experiment. 10^9 B-cells receptors sequenced,
 10^4 antibody-antigen interaction measured



Lineages that bind the antigen have unique properties.

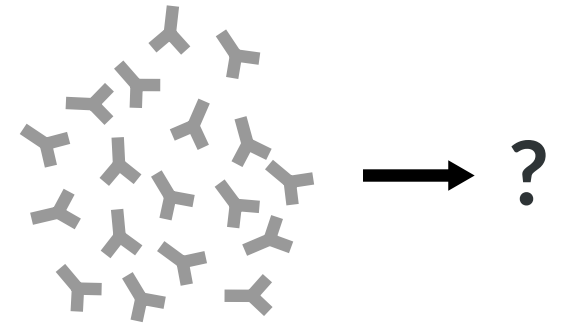
Summary

Vaccination experiment. 10^9 B-cells receptors sequenced,
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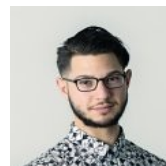
Lineages that bind the antigen have unique properties.

In-repertoire replicates : the same
receptors in different individuals follow
the same patterns.



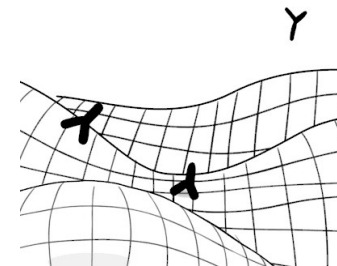
THANKS !

Tatiana Ruiz-Bedoya
Giulio Isacchini
Angela Phillips
Zach Montague
Natanael Spisak
Katherine Lawrence
Michael Desai
Alexandra Poret
Alief Moulana
Thierry Mora
Daniel Newton
Joao Ascensao
Berk Alpay
Aleksandra Walczak



TD Lab

Virginie Chatain
Charlotte Réocreux

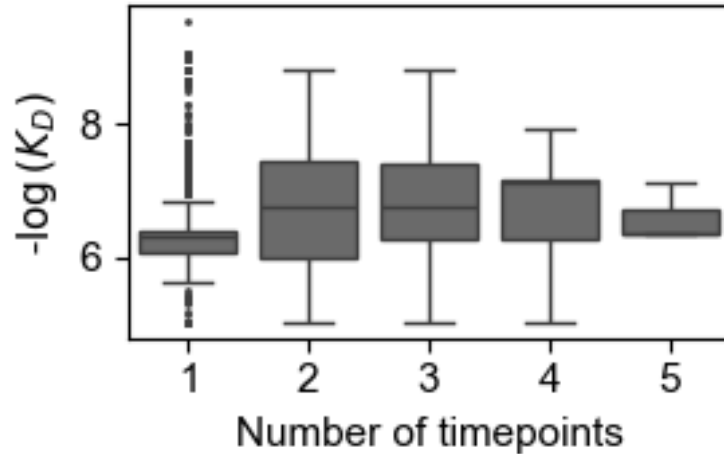


We're looking for a post-doc!

dupic@ciml.univ-mrs.fr

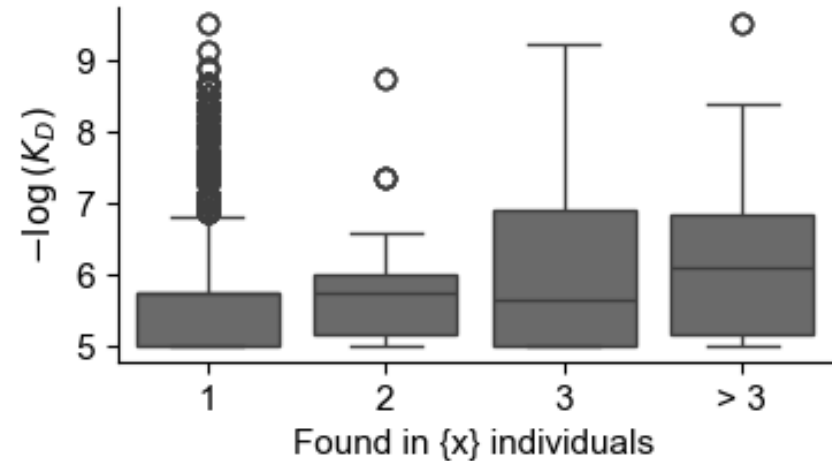
www.thopic.eu

The affinity of the germline sequence matters too !

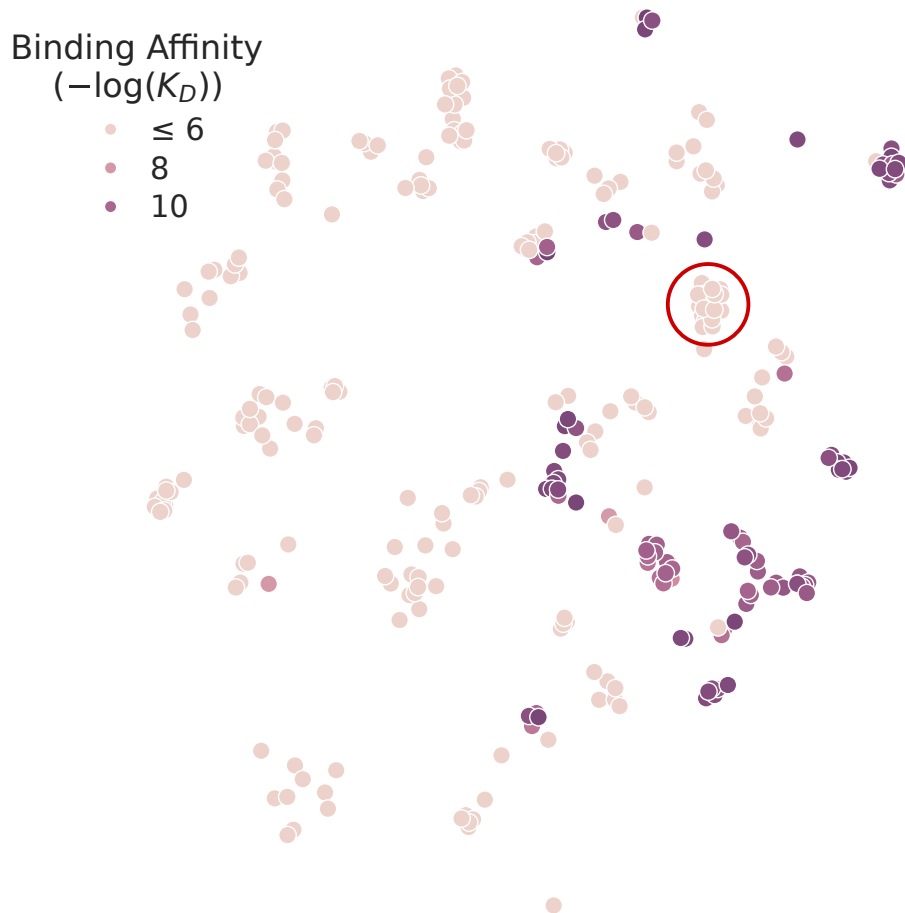


High affinity germline
are found across more
timepoints

High affinity germline
are more shared



Structural properties



AlphaFold is not doing very well here but is showing promise.

UMAP based on AlphaFold internal representation. Circled sequences are potential predicted binders.



**We are recruiting up to
two new team leaders.**

Advancing Immunology Together

200 scientists • 16 research teams

State-of-the-art facilities:

Flow cytometry • Imaging • Genomics •
Histology • Computational biology • CRISPR
screening

Start-up package:

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funding • Free core facilities

Application deadline: May 1st 2026

Contact: milpied@ciml.univ-mrs.fr
<https://ciml.univ-mrs.fr>